

A RELATIONAL THEORY OF POWER ALTERNATION

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Abstract

A long-standing thesis in political economy relates cooperation among rival elites to mechanisms for power alternation. What remains unclear is why ruling elites would voluntarily transfer power if they can instead sustain cooperation by granting political concessions. To resolve this puzzle, we study a setting without external enforcement in which two elite groups may cooperate repeatedly, and the party in power can both make concessions and transfer power. Incumbents cede power in equilibrium because power alternation disciplines concessions through continuation values under future incumbencies, rather than inefficient punishments such as breakdowns of cooperation. Efficient self-enforcing arrangements imply norms that maximize the stability of power, with alternation occurring only when necessary to sustain cooperation. Finally, as the cost of transferring power increases, alternation becomes more frequent and power stability deteriorates.

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1. INTRODUCTION

Stable political order requires cooperation among rival elites. *Power alternation* has long been viewed as the key to sustaining elite cooperation. [Pareto \(1935\)](#) distinguished governing and nongoverning elites and emphasized the importance of their alternation for long-term prosperity. [Schumpeter \(1942\)](#) stressed the link between institutional arrangements of power alternation and elite cooperation, writing that “the democratic processes reproduce itself steadily without creating situations that enforce resort to non-democratic methods and...it copes with current problems in a way which all interests that count politically find acceptable in the long run” (p. 290). [Przeworski \(1999\)](#) further articulated this insight, contending that power alternation constitutes a “minimalist” definition of democracy as it induces conflicting political forces to “comply with the rules rather than engage in violence” (p. 45). The effect of power alternation on inducing cooperative outcomes has been well established in existing works ([Alesina, 1988](#); [Dixit, Grossman and Gul, 2000](#); [Acemoglu, Golosov and Tsyvinski, 2011](#); [Egorov and Sonin, 2015](#); [Helmke, Kroeger and Paine, 2022](#); [Invernizzi and Ting, 2024](#); [Delgado-Vega and Schneider, 2024](#); [Delgado-Vega, 2025](#)). Yet these works typically treat power alternation as exogenously enforced, leaving open why ruling elites would voluntarily relinquish power.

Power alternation is not the only mechanism to sustain elite cooperation. [Kelsen \(1929\)](#) proposed a competing argument and emphasized that governing elites could instead secure cooperation of nongoverning elites through *political concessions*, without relinquishing power ([Przeworski, 2010](#)). The effect of concessions in buying off rivals has been central to the literature of authoritarian power sharing ([Gandhi and Przeworski, 2007](#); [Gandhi, 2008](#); [Svolik, 2012](#); [Powell, 2013](#); [Meng, Paine and Powell, 2023](#); [Egorov and Sonin, 2024](#)). Compared to granting concessions, transferring power is inherently costly and risky for governing elites, involving the cost of organizing a power transition, the loss of established patronage networks, and the uncertainty after losing power ([Fearon, 2011](#); [Egorov and Sonin, 2015](#)).¹

The puzzle, therefore, is why elites would transfer power to maintain cooperation, given the costs and risks of relinquishing power and the availability of political concessions. To address this puzzle, we study self-enforcing arrangements in which governing elites can both grant concessions and transfer power, and characterize how these instruments are traded off in sustaining cooperation absent external enforcement. Our analysis naturally combines

¹[Fearon \(2011\)](#) emphasizes that ruling elites may prefer concessions to power transfer when the latter is costly and risky. [Egorov and Sonin \(2015\)](#) model environments in which power alternation is exogenous and losing power entails existential risks for incumbents, underscoring that relinquishing power can be both costly and dangerous.

themes from political economy and organizational economics (Kreps, 1990; Baker, Gibbons and Murphy, 2002; Levin, 2003). In particular, both concessions and power alternation must be supported by relational incentives among elites. We show that power alternation becomes necessary because it disciplines concessions without inefficient breakdowns of cooperation.

That the foundations of political systems must be self-enforcing has been emphasized in political economy (Weingast, 1997; Grief, 1998; Myerson, 2008; Fearon, 2011). Indeed, scholars have documented informal pacts of elite power alternation in autocracies that lack strong institutional enforcement—for instance, in Colombia’s National Front period (Bushnell, 1993), in Mexico under the Institutional Revolutionary Party (Magaloni, 2006), and in China between major factions within the Chinese Communist Party (Nathan, 1973; Shirk, 1993; Francois, Trebbi and Xiao, 2023).² Moreover, historical development of democracy suggests that alternation of power had been practiced before the introduction of formal institutions. Prominent examples include 18th-century Britain and Sweden, where political power peacefully alternated back and forth between two conflicting elite factions—the Whigs and the Tories in Britain (Plumb, 1967; Brewer, 1976), and the Hats and the Caps in Sweden (Roberts, 2003)—as conventions. The corresponding political arrangement was described as “competitive oligarchy” by Dahl (1971), who argued that it could first develop spontaneously among elites and create a “viable system of mutual security,” laying the foundation for later institutionalized mass democracy by allowing new social strata to be “more easily socialized into the norms and practices of competitive politics already developed among the elites.” (Dahl, 1971, p. 36).³

In this paper, we consider an environment without any external enforcement institutions, where two parties (elite groups, or political factions) interact and may cooperate repeatedly, while the party in power could make concessions as well as transfer power to the other party. At the beginning of every period, power resides with one party, which we refer to as the *incumbent*, while the other party is the *opposition*. Each party first chooses whether to cooperate or to take its outside option. If both cooperate, a state of the world—good or bad—is then realized. Resources are available only in the good state. Holding power, the incumbent privately observes the realization of the state and, if favorable, allocates resources between the two parties, thereby making political concessions through resource sharing. Out of power, the opposition is uninformed about the state and observes only the

²Bushnell (1993) documents that the Colombian National Front (1958-1974) served as an elite agreement ensuring alternation between the Liberal and Conservative parties to end partisan violence. Magaloni (2006) interprets the Institutional Revolutionary Party in Mexico as an elite pact maintained through regularized presidential succession. Nathan (1973) describes alternating dominance among party factions in Chinese factional politics.

³For more on how stable democratization was driven by the preexisting elite alternation in Britain and Sweden, see Cowling (1967), Ansell and Samuels (2014), Ziblatt (2017), and Rustow (1955).

amount of resources shared by the incumbent. At the end of the period, the incumbent chooses whether to transfer power at a cost. If power transitions, the current opposition becomes the new incumbent in the next period.

Critical to our analysis are three driving forces. First, to cooperate, the two parties must forsake their outside options. Hence, the sustainability of cooperation relies on their expectations over the exercise of power through the allocation of resources within a period and the distribution of power through power transfer across periods. Second, power conveys both control over resources that can be used for political concessions and superior information about the availability of these resources. As a result, the incumbent’s ability to promise concessions is hindered by two frictions: the incumbent’s commitment problem and information asymmetry. Third, relinquishing power is costly to its holder and must be voluntary. Therefore, the transfer of power must be compensated for by how power will be exercised and distributed under the new incumbency formed after the transition.

We address the puzzle by characterizing the interplay among power alternation, political concessions, and sustained cooperation. We begin by formalizing [Schumpeter’s \(1942\)](#) intuition: power alternation alone can be self-enforcing and sustain cooperation through intertemporal redistribution of resources. However, such equilibria are Pareto suboptimal, as costly power transfers must occur frequently in the absence of concessions. Conversely, although concessions do not directly reduce surplus, they are insufficient for sustained cooperation under the two frictions. Because the opposition cannot distinguish a bad state from the incumbent’s expropriation, disciplining concessions requires on-path punishments that eventually lead to a breakdown of cooperation.⁴

Our first main result establishes that an equilibrium with sustained cooperation exists if and only if parties are sufficiently patient and the cost of power transfer is not too high. More importantly, whenever feasible, every efficient equilibrium exhibits sustained cooperation and must combine political concessions with power alternation. Concessions reduce the frequency of costly power transfers by reallocating resources within periods, while power alternation ensures the credibility of concessions: incumbents who repeatedly fail to make concessions relinquish power, and cooperation continues under the new incumbency.

We thus identify a novel rationale for power alternation: beyond intertemporal redistribution, it *disciplines concessions while preserving the continuation of cooperation*.

⁴In Appendix C, we assume that the opposition has additional information captured by a signal generated by the state. We show that this result stills holds whenever this signal fails to perfectly reveal the true realization of the state. Hence, concessions alone are insufficient for sustained cooperation so long as the opposition cannot perfectly distinguish between a bad state and the incumbent’s expropriation in a good state. Broadly, this result is reminiscent of the immiseration result typical of dynamic settings with private information. For the similar results in different settings, see [Thomas and Worrall \(1990\)](#), [Atkeson and Lucas \(1992\)](#), [Yared \(2010\)](#), [Li and Matouschek \(2013\)](#), and [Lipnowski and Ramos \(2020\)](#).

Crucially, because power transfer is voluntary and must be compensated for, discipline *does not* operate through the cost of relinquishing power. Instead, it stems from how power is exercised and distributed after transitions. Hence, it is the relationship under the future incumbency that enforces the relationship under the current incumbency.⁵ Our results therefore contrast with models where the provision of dynamic incentives under information asymmetry relies on long-run inefficiencies (for instance, [Green and Porter, 1984](#); [Padró-i Miquel and Yared, 2012](#); see also [Yared, 2010](#); [Li and Matouschek, 2013](#); [Lipnowski and Ramos, 2020](#)). We uncover a distinct political economy force to avoid such inefficiencies: although information asymmetry pervades the exercise of power, rival elites can endogenously reallocate control to sustain credible concessions and continued cooperation.

Our second set of results elucidates how such discipline works by characterizing the exercise and distribution of power in efficient self-enforcing arrangements. We derive a sharp geometric characterization for the sets of the equilibrium *interim* continuation values that the parties expect after resource sharing and before power transfer. This enables us to determine whether continuation values are attained with or without power transfer and identify three norms of power alternation in all efficient equilibria.

The first norm governs *when* power alternation should occur and illustrates how power transfer and political concessions are traded off throughout each incumbency in an efficient equilibrium. Along the equilibrium path, every incumbent retains power if and only if its continuation value exceeds its minimal cooperative payoff.⁶ For a given continuation value, the equilibrium prioritizes preserving the current incumbency through prescribing future political concessions, and prescribes power transfer in the current period only when this value becomes insufficient to sustain cooperation under the same incumbency. We refer to this property as the *norm of backloaded abdication*: power transfer is delayed until it becomes indispensable for the continuation of cooperation.

The second norm governs *how* power transfer should occur. Whenever power is transferred, the incumbent is indifferent between relinquishing power and then continuing cooperation under the new incumbency, and retaining power and then taking its outside option thereafter. Hence, in every continuation equilibrium immediately after a power transition, the future surplus is divided to maximize the new incumbent's continuation value, with the remainder just compensating the previous incumbent for relinquishing power. We refer to this property as the *norm of almighty inauguration*, which can be

⁵Hence, power alternation can assure the credibility of political concessions in efficient equilibria no matter how small the cost of transferring power is. Furthermore, we show that when this cost is sufficiently small, every efficient equilibrium attains sequential efficiency on the path of play. See [Section 3](#).

⁶Specifically, in any equilibrium, given that cooperation is prescribed, the incumbent can secure a minimal cooperative payoff from sharing no resources in the current period and taking its outside option thereafter.

interpreted as an implicit bargaining protocol for the play after power transition that assigns maximal bargaining leverage to the inaugurating party.

The third norm governs behaviors within each incumbency. While the norm of backloaded abdication implies that an incumbent’s continuation value determines its ability to preserve power through political concessions, the norm of almighty inauguration implies that this ability is maximized for the new incumbent after each power transition. Beyond these two norms, any efficient equilibrium further features a *norm of minimal compromise*: in any continuation play within an incumbency, the incumbent’s continuation value—and thus its capacity to preserve power through concessions—is maximized.

Together, these norms characterize how power should be efficiently exercised and distributed. They imply maximal power stability: incumbencies persist whenever feasible and power alternation occurs only when necessary to sustain cooperation. This perspective provides a new link between the stability of power and the sustainability of cooperation.⁷

Our final set of results characterizes the on-path behaviors in any efficient equilibrium, which allows us to derive comparative statics on how the equilibrium stability of power varies with the cost of transferring power. We define an incumbency as *hegemonic* if the incumbent faces no immediate risk of losing power. Three types of political dynamics may emerge. When the cost is high, hegemony is unattainable, and the political dynamic is *unstable*, with frequent power transitions. When the cost is low, every incumbency begins as hegemonic, leading to a *stable* political dynamic with enduring incumbencies. When the cost is intermediate, incumbencies are initially precarious but become hegemonic following sufficiently many favorable state realizations. In this case, a *stabilizable* political dynamic arises, featuring cycles between stable phases with persistent power and unstable phases with frequent power transitions.

Surprisingly, as the cost of transferring power increases, transitions become more frequent and thus power stability deteriorates. This result rests on a crucial driving force of our analysis: transfer of power is voluntary and must be compensated for by the exercise and

⁷The existing literature in political economy provides contrasting views on this relation. On the one hand, some papers show that the risk of losing power undermines the value of future interactions for the governing group, and thus advance the intuition that power stability facilitates cooperation between groups in and out of power. See, for instance, [Olson \(1993\)](#), [McGuire and Olson \(1996\)](#), and [Acemoglu \(2003\)](#). However, all of these papers assume a fixed power structure without considering the possibility of power alternation among groups. On the other hand, another set of works shows that the prospect of power switches enables dynamic incentives that induce every governing group to take cooperative actions. See, for instance, [Alesina \(1988\)](#), [Dixit, Grossman and Gul \(2000\)](#), [Acemoglu, Golosov and Tsyvinski \(2011\)](#), and [Egorov and Sonin \(2015\)](#). In particular, [Dixit, Grossman and Gul \(2000\)](#) and [Acemoglu, Golosov and Tsyvinski \(2011\)](#) make the point that the stability of power in fact weakens the discipline for cooperative actions. However, all of these papers assume exogenous power fluctuations, and do not consider the inherent costs and risks of power transfer for a governing group.

distribution of power after the transition. As the cost rises, more generous compromises are required throughout every incumbency to motivate power transfers. This, however, undermines every incumbent’s ability to retain power, thereby leading to more frequent power transitions.

The remainder of the paper is organized as follows. Section 2 describes the model. Section 3 analyzes the interplay among power alternation, political concessions, and sustained cooperation. Section 4 characterizes the norms of power alternation. Section 5 presents the dynamics of behaviors along with comparative statics on power stability. Section 6 situates our results in historical contexts. Finally, Section 7 concludes. The proofs of the main results are in Appendix A; omitted formal statements and extensions are in Appendix B and C.

Related literature

This paper investigates the interplay among power alternation, political concessions, and sustained cooperation, and explores how this interplay shapes the distribution of power in an environment stripped of any external enforcement. Our paper is thus mostly related to the literature on the role of power fluctuations in various political economy environments (Alesina, 1988; Dixit, Grossman and Gul, 2000; Acemoglu, Golosov and Tsyvinski, 2011; Aguiar and Amador, 2011; Egorov and Sonin, 2015; Helmke, Kroeger and Paine, 2022; Invernizzi and Ting, 2024; Delgado-Vega and Schneider, 2024; Delgado-Vega, 2025). Much of the literature assume some exogenous process of power switches and study how it supports cooperative behavior.⁸ An important precursor is Dixit, Grossman and Gul (2000). Our approach differs along three key dimensions. First, they study how political concessions can be supported by exogenous power alternation in a stochastic game with perfect monitoring, whereas we model the options of cooperating, offering concessions, and transferring power explicitly, and focus on the interplay among them in self-enforcing arrangements. Second, in their model, power alternation supports concessions by enabling off-path continuation punishments: after a deviation, no subsequent party in power would make concessions. In contrast, in our model, power alternation enables on-path continuation punishments under continued cooperation: concessions are enforced by the exercise and distribution of power after transitions. Furthermore, how this discipline works is endogenously determined and characterized by the norms of power alternation.

Our work also joins the political economy literature on the endogenous persistence and transitions of power (Acemoglu and Robinson, 2001; Gomes and Jehiel, 2005; Acemoglu

⁸Exceptions are Delgado-Vega and Schneider (2024) and Delgado-Vega (2025). They partially endogenize power alternation in relational contracting environments by considering models in which the support of a long-lived third player influences the evolution of power between two parties.

and Robinson, 2008; Acemoglu, Egorov and Sonin, 2008, 2012, 2015, 2018; Bidner, Francois and Trebbi, 2014; Diermeier, Egorov and Sonin, 2017). An overwhelming majority of the literature model power as state variables of dynamic games and focus on Markovian strategies. We depart from them by exploring how power dynamics among elite groups endogenously arise as part of relational incentives and can exhibit history dependence.

Our approach builds on the literature of dynamic contracting under information asymmetry (Yared, 2010; Li and Matouschek, 2013; Abdulkadiroglu and Bagwell, 2013; Ales, Maziero and Yared, 2014; Fahn and Klein, 2019; Lipnowski and Ramos, 2020; Li, Mukherjee and Vasconcelos, 2023; Acharya, Lipnowski and Ramos, 2025; Gieczewski and Li, 2024; Hörner and Sanktjohanser, 2025). Closest to our model are Yared (2010) and Li and Matouschek (2013). Both papers consider a model in which one player privately observes the feasibility of concessions made to the other player in every period, and find dynamics analogous to the within-incumbency behavior of efficient equilibria in our model. Despite these common threads, the novel feature of our paper is that parties can endogenously reallocate control over resources through power transfer, which is central in political economy, but has not been investigated from the perspective of dynamic agency relationships. The option to transfer power fundamentally changes the nature of incentive provision. In their papers, on-path punishments to enforce concessions are derived from inefficiencies in the long run—war in Yared (2010) and decline in effort provision in Li and Matouschek (2013).⁹ Whereas in our model, on-path punishments entail power transitions rather than the breakdown of cooperation. Concessions under one incumbency are enforced by the relationship under the future incumbency formed after a transition.

Finally, our focus on power alternation relates to the relational contracting models studying the dynamics of decision rights (Baker, Gibbons and Murphy, 1999; Li, Matouschek and Powell, 2017), the optimal structure of control rights (Baker, Gibbons and Murphy, 2002; Rayo, 2007; Ferreira and Li, 2025), and renegotiation and bargaining (Miller and Watson, 2013; Halac, 2015; Safronov and Strulovici, 2018; Watson, Miller and Olsen, 2020). We complement these papers by allowing the parties to explicitly transfer power in every period, characterizing the equilibrium distribution of power, and showing how efficient self-enforcing agreements involve an implicit bargaining protocol for the continuation play after power transition.

⁹Ales, Maziero and Yared (2014) and Acharya, Lipnowski and Ramos (2025) also feature similar dynamics, focusing on political turnover rather than power alternation, with incumbents replaced after consecutive adverse shocks. A few other papers analyze models of dynamic moral hazard, where the principal has private information about either her persistent type (Halac, 2012) or the agent’s performance (Levin, 2003; Fuchs, 2007), and show that optimal relational contracts necessitate on-path termination.

2. MODEL

The players are two *parties* (“political factions,” or “elite groups”), $j \in \{0, 1\}$, interacting at $t \in \{0, 1, 2, \dots\}$.

Economy. At each t , each party j chooses whether to *cooperate*, $c_{j,t} \in \{0, 1\}$. Cooperation fails if at least one party chooses not to cooperate, $c_{0,t}c_{1,t} = 0$, in which case each j gets a reservation value $\underline{u}_j > 0$ from its outside option. If both parties choose to cooperate, $c_{0,t}c_{1,t} = 1$, a state of the world $\theta_t \in \{0, 1\}$ is realized. In the good state $\theta_t = 1$, resources accrue to the two parties as a whole, with the total amount normalized to 1; while in the bad state $\theta_t = 0$, no resources are available. The state is stochastically drawn from a stationary distribution with $\Pr(\theta_t = 1) = q \in (0, 1)$. Throughout, we assume

Assumption 1. $q > \underline{u}_0 + \underline{u}_1$.

Under this assumption, it is efficient for the two parties to always cooperate. As long as resources are properly allocated in good states, they are willing to cooperate. The allocation of resources, however, is a matter of politics.¹⁰

Politics. One party, $I_t \in \{0, 1\}$, holds *power* at t and is referred to as the *incumbent*. The other party, $1 - I_t$ is out of power and referred to as the *opposition*.

Holding power, the incumbent is able to observe the state of the world θ_t and, if the realization is good, $\theta_t = 1$, it chooses how to allocate resources between the two parties, sharing $x_t \in [0, 1]$ with the opposition and preserving the remaining $1 - x_t$. The opposition knows the amount of resources shared by the incumbent, $s_t := \theta_t x_t$. Being out of power, however, the opposition has difficulty observing the state of the world θ_t . Therefore, when it receives no resources, $s_t = 0$, the opposition is uncertain about whether the state was bad, $\theta_t = 0$, or whether the state was good, $\theta_t = 1$, but the incumbent has expropriated all resources, $x_t = 0$.

At the end of t , the incumbent has the option to transfer its power to the opposition, $a_t \in \{0, 1\}$. If $a_t = 1$, power transitions and the current opposition becomes the new incumbent at $t + 1$, $I_{t+1} = 1 - I_t$. Otherwise, if the incumbent chooses to hold onto power, $a_t = 0$, its power persists to $t + 1$, $I_{t+1} = I_t$. Transferring power imposes on the incumbent a cost $k > 0$.

Timing and payoffs. The timing of the stage game at each t is as follows.

1. Each party j publicly and simultaneously chooses whether to cooperate, $c_{j,t} \in \{0, 1\}$.
2. If $c_{0,t}c_{1,t} = 1$, the state of the world $\theta_t \in \{0, 1\}$ is realized.

¹⁰The game is trivial when this assumption fails: cooperation is impossible because no matter how the resources are allocated, at least one party prefers its outside option.

3. The incumbent I_t privately observes θ_t and, if $\theta_t = 1$, it allocates resources, $x_t \in [0, 1]$.
4. The opposition $1 - I_t$ observes the amount of resources it receives, $s_t = \theta_t x_t$.
5. The incumbent I_t publicly chooses whether to transfer power, $a_t \in \{0, 1\}$.

The initial incumbent, $I_0 \in \{0, 1\}$, is exogenously given and the incumbent at each $t \geq 1$ is endogenously determined via

$$I_t = (1 - c_{0,t-1}c_{1,t-1}a_{t-1})I_{t-1} + c_{0,t-1}c_{1,t-1}a_{t-1}(1 - I_{t-1}).$$

The flow payoff each j receives at t is

$$c_{j,t}c_{1-j,t}y_{j,t} + (1 - c_{j,t}c_{1-j,t})\underline{u}_j,$$

where

$$y_{j,t} := \begin{cases} \theta_t(1 - x_t) - a_t k, & I_t = j \\ \theta_t x_t, & I_t = 1 - j \end{cases}$$

is its flow payoff under cooperation. Both parties are forward looking and share a common discount factor $\delta \in (0, 1)$. All payoffs are normalized as per period averages.

Interpretations. Our framework is designed to understand the interplay among elite cooperation, political concessions, and power alternation in an environment stripped of external enforcement institutions. It consists of three driving forces.

First, to generate resources, both parties must forsake their outside options to cooperate. Resources capture available rents, spoils, and feasible policies in a regime, which are possible only when both rival elite groups participate in the regime and contribute to its survival ($c_{0,t}c_{1,t} = 1$) rather than act independently or even engage in conflict ($c_{0,t}c_{1,t} = 0$). The sustainability of cooperation relies on the expectations over how the stream of resources will be allocated, which is determined by the exercise and distribution of power.

Second, an incumbent can exercise power to make political concessions, which may take the form of spoil sharing or policy compromises (Gandhi, 2008).¹¹ We model these concessions as the allocation of resources within a period (x_t). However, frictions inherent in the exercise of power impedes the credibility of these concessions: power conveys both control over resources and superior information about the availability of these resources. In

¹¹Prominent examples of systematic grant of political concessions by governing elites include the use of patronage by Robert Walpole—the first Whig prime minister in the 18th-century Britain—to maintain Whig hegemony (Plumb, 1956b), and *trasformismo*—systematic practices of compromises to absorb contesting political forces—in the 19th-century Italy after its unification (Valibruzzi, 2015).

particular, the exercise of power involves evaluating and making policies under complex and subtle conditions, and thus is inevitably imperfectly monitored by those outside the government. As a result, the group out of power cannot always verify the excuses that the group in power uses to avoid making concessions. That promises of concessions are hindered by information asymmetry has been stressed in political economy. However, many papers explore how specific formal institutions can be maintained to eliminate this friction—for instance, court in Myerson (2008, 2015), regular elections in Fearon (2011), and parliament in Boix and Svolik (2013). In contrast, our focus is to understand how political concessions and power transfers are supported and traded off absent any formal institutions. Hence, this friction remains one of the driving forces of our analysis. We capture it in a stark way: only the incumbent privately observes the state of the world θ_t . Nevertheless, central to our analysis is only that the opposition *cannot perfectly* distinguish between a bad state realization and the incumbent’s expropriation. We return to this issue after Proposition 2.

Finally, the distribution of power depends on transfer of power across periods. Relinquishing power ($a_t = 1$) is costly, involving the cost of organizing a power transition, the loss of established patronage network, and the uncertainty of losing power (Egorov and Sonin, 2015). We model these costs in a reduced form by assuming that transferring power imposes a cost k on the incumbent. While other modeling approaches are equivalent, our assumption sharpens the focus on one critical driving force of our analysis: power transfer is voluntary and must be compensated for by how power will be exercised and distributed after the transition.

Strategies and equilibria. We assume a public randomization device. At each t , immediately after the opposition observes the amount of resources shared by the incumbent, both parties observe the realization of z_t , drawn from the uniform distribution on $[0, 1]$. Conditional on z_t , the incumbent may randomize its power transfer decision at t and both parties may randomize their subsequent decisions at $t + 1$.¹²

The solution concept is *perfect public equilibrium* (equilibrium). The two parties can condition their decisions on the public history of play: $h^0 := I_0$ and, for each $t \geq 1$,

$$h^t := (h^{t-1}, c_{0,t-1}, c_{1,t-1}, s_{t-1}, z_{t-1}, a_{t-1}, I_t)$$

records all publicly observed outcomes up to the beginning of t . The set of all possible histories is denoted as $\mathcal{H}$. A strategy profile σ specifies three decisions for each j , $(C_j^\sigma, X_j^\sigma, A_j^\sigma)$. For each $h^t \in \mathcal{H}$, first, $c_{j,t} = C_j^\sigma(h^t) \in \{0, 1\}$ is j ’s decision on whether to

¹²The existence of a public randomization device is a common assumption in the literature of repeated games and is used to convexify the sets of equilibrium continuation values.

cooperate; second, $x_t = X_j^\sigma(h^t) \in [0, 1]$ is j 's resources allocation decision given that j holds power, $I_t = j$, that the both parties cooperate, $c_{0,t}c_{1,t} = 1$, and that the state of the world is good, $\theta_t = 1$; third, $a_t = A_j^\sigma(s_t, z_t|h^t) \in \{0, 1\}$ is j 's decision on whether to transfer power given that j currently holds power, $I_t = j$, that both parties cooperate, $c_{0,t}c_{1,t} = 1$, the amount of resources the opposition $1 - j$ receives, s_t , and the realization of z_t . At the beginning of each t , given the history h^t , σ induces for each j the payoff of

$$U_j(\sigma|h^t) := \mathbb{E}^\sigma \left((1 - \delta) \sum_{\tau=t}^{\infty} \delta^{\tau-t} (c_{j,t}c_{1-j,t}y_{j,t} + (1 - c_{j,t}c_{1-j,t})\underline{u}_j) \mid h^t \right).$$

A strategy profile is an *equilibrium* if following every possible history, each decision it prescribes for each party is optimal for that party given all the other decisions. An equilibrium is *efficient* if it maximizes the total surplus between the two parties at the beginning of $t = 0$ among all possible equilibria.¹³ Formally, let Σ denote the set of all equilibria and $\Sigma^* \subseteq \Sigma$ denote the set of all efficient equilibria. First, $\sigma \in \Sigma$ if for each $h^t \in \mathcal{H}$ and each $j \in \{0, 1\}$,

$$C_j^\sigma(h^t) \in \arg \max_{c_{j,t} \in \{0,1\}} \mathbb{E}^\sigma \left((1 - \delta) (c_{j,t}c_{1-j,t}y_{j,t} + (1 - c_{j,t}c_{1-j,t})\underline{u}_j) + \delta U_j(\sigma|h^{t+1}) \mid h^t, c_{j,t} \right);$$

and

$$X_j^\sigma(h^t) \in \arg \max_{x_t \in [0,1]} \mathbb{E}^\sigma \left((1 - \delta)(1 - x_t - a_t k) + \delta U_j(\sigma|h^{t+1}) \mid h^t, c_{0,t}c_{1,t} = 1, s_t = x_t \right);$$

given that $I_t = j$, $c_{0,t}c_{1,t} = 1$, and $\theta_t = 1$; and

$$A_j^\sigma(s_t, z_t|h^t) \in \arg \max_{a_t \in \{0,1\}} \mathbb{E}^\sigma \left(-(1 - \delta)a_t k + \delta U_j(\sigma|h^{t+1}) \mid h^t, c_{0,t}c_{1,t} = 1, s_t, z_t, a_t \right)$$

holds for all $s_t, z_t \in [0, 1]$, given that $I_t = j$ and $c_{0,t}c_{1,t} = 1$. Second, $\sigma \in \Sigma^*$ if

$$U_0(\sigma|I_0) + U_1(\sigma|I_0) \geq U_0(\sigma'|I_0) + U_1(\sigma'|I_0)$$

holds for all $\sigma' \in \Sigma$.

¹³Here, efficiency is narrowly defined for the two parties as a whole and does not necessarily imply the maximization of social welfare. Rather, the two parties could collude for extractive purposes against the welfare of the general public. In fact, one of our main results shows that efficiency is equivalent with maximizing the payoff of the initial incumbent, see Corollary 4.

3. ALTERNATION, CONCESSIONS, AND COOPERATION

We are particularly interested in whether the two parties can sustain cooperation and whether incumbents would transfer power to oppositions.

Definition 1. An equilibrium $\sigma \in \Sigma$ has

1. *sustained cooperation* if on the equilibrium path of σ , $c_{0,t}c_{1,t} = 1$ holds for all t ;
2. *power alternation* if on the equilibrium path of σ , for each $t \geq 0$, there exists a $\tau \geq t$ such that $a_\tau = 1$ and $I_{\tau+1} = 1 - I_\tau$.

An equilibrium attains sustained cooperation if parties always cooperate with each other. An equilibrium exhibits power alternation if no party monopolizes power: whenever a party holds power, it eventually relinquishes it to the other. In such an equilibrium, power alternates over time between the two parties.

In this section, we first investigate Schumpeter’s insight by analyzing a class of equilibria in which incumbents make no political concessions. We then identify the constraints required for cooperation and show that they reflect two frictions—the incumbent’s commitment problem and information asymmetry—that limit an incumbent’s ability to promise concessions. These frictions imply that political concessions alone are insufficient, making power alternation necessary for sustaining cooperation. We next analyze how the option to transfer power expands the set of attainable continuation values. Finally, we derive the necessary and sufficient conditions for sustained cooperation and characterize the interplay among power alternation, political concessions, and sustained cooperation.

3.1. Schumpeterian equilibria

Schumpeter (1942) argued that parties cooperate with each other because they expect power to alternate as long as cooperation proceeds.¹⁴

Definition 2. An equilibrium $\sigma \in \Sigma$ is *Schumpeterian* if on its equilibrium path, $x_t = 0$ holds for all t .

In a Schumpeterian equilibrium, each party, when holding power, shares no resources with the opposition. Such an equilibrium reflects Schumpeter’s insight: the opposition is

¹⁴Przeworski (1999) elaborates Schumpeter’s idea as follows: “[T]he very prospect that governments would alternate may induce the conflicting political forces to comply with the rules rather than engage in violence...Bloodshed is avoided by the mere fact that, à la Aristotle, the political forces expect to take turns.”

never compensated *within a period* through resource sharing; instead, it is compensated *intertemporally* with the prospect of assuming power itself in the future.

Proposition 1. *There exists a Schumpeterian equilibrium with sustained cooperation if and only if $\delta > \delta^*$ and $k \leq \underline{k}$, where*

$$\delta^* := \max \left\{ \frac{\underline{u}_0}{q - \underline{u}_1}, \frac{\underline{u}_1}{q - \underline{u}_0} \right\}$$

and $\underline{k} > 0$. Moreover, for every Schumpeterian equilibrium, there exists another equilibrium that Pareto dominates it.

The above proposition confirms Schumpeter's intuition. As long as the two parties are sufficiently patient and the cost of transferring power is sufficiently low, sustained cooperation is achieved in at least one Schumpeterian equilibrium. Through alternating incumbencies, the two parties divide resources across periods, so as to keep both interested in continuing cooperating with each other.¹⁵

However, Proposition 1 also implies that Schumpeterian equilibria are Pareto suboptimal and thus inefficient. Power transitions occur too frequently, and each incumbent bears the cost of relinquishing power at every transition. Relative to such equilibria, an incumbent can benefit from reducing the frequency of power transfer while offsetting the opposition's loss through resource sharing. Hence, the two parties may cooperate more efficiently when political concessions are incorporated.

3.2. Insufficiency of concessions and necessity of alternation

The incumbent's incentive to share resources is governed by its continuation value given the amount of resources the opposition receives. Formally, let

$$\Gamma_j^\sigma(s_t|h^t) := \begin{cases} \mathbb{E}^\sigma \left(U_j(\sigma|h^{t+1}) - \frac{1-\delta}{\delta} a_t k \mid h^t, c_{0,t}c_{1,t} = 1, s_t \right), & I_t = j \\ \mathbb{E}^\sigma \left(U_j(\sigma|h^{t+1}) \mid h^t, c_{0,t}c_{1,t} = 1, s_t \right), & I_t = 1 - j \end{cases}$$

be party j 's continuation value under strategy profile σ given that following history h^t both parties have chosen to cooperate, $c_{0,t}c_{1,t} = 1$, and the opposition has received s_t . This is the *interim* continuation value party j expects right after observing the realization of s_t and before the realization of the public randomization device z_t .

¹⁵In particular, for each j , because party j 's payoff at the beginning of its incumbency is bounded below $q - \underline{u}_{1-j}$, in any Schumpeterian equilibrium with sustained cooperation, party j must have a positive probability to transfer power after no more than $\max \{t : \delta^t \geq \underline{u}_{1-j}/(q - \underline{u}_j)\}$ periods in power.

Lemma 1. *If $\sigma \in \Sigma$, $h^t \in \mathcal{H}$ is on the equilibrium path of σ , and $C_0^\sigma(h^t)C_1^\sigma(h^t) = 1$, then*

$$\begin{aligned}
(\text{IR-I}) \quad & U_{I_t}(\sigma|h^t) \geq \underline{u}_{I_t} \\
(\text{IR-O}) \quad & U_{1-I_t}(\sigma|h^t) \geq \underline{u}_{1-I_t} \\
(\text{PK-I}) \quad & U_{I_t}(\sigma|h^t) = (1-q)\delta\Gamma_{I_t}^\sigma(0|h^t) + q\left((1-\delta)\left(1-X_{I_t}^\sigma(h^t)\right) + \delta\Gamma_{I_t}^\sigma\left(X_{I_t}^\sigma(h^t) \mid h^t\right)\right) \\
(\text{PK-O}) \quad & U_{1-I_t}(\sigma|h^t) = (1-q)\delta\Gamma_{1-I_t}^\sigma(0|h^t) + q\left((1-\delta)X_{I_t}^\sigma(h^t) + \delta\Gamma_{1-I_t}^\sigma\left(X_{I_t}^\sigma(h^t) \mid h^t\right)\right) \\
(\text{IC}) \quad & (1-\delta)\left(X_{I_t}^\sigma(h^t) - x_t\right) \leq \delta\left(\Gamma_{I_t}^\sigma\left(X_{I_t}^\sigma(h^t) \mid h^t\right) - \Gamma_{I_t}^\sigma(x_t|h^t)\right), \forall x_t \in [0, 1].
\end{aligned}$$

Lemma 1 shows the constraints necessary for the two parties to cooperate in a given period. Conditions (IR-I) and (IR-O) reflect that cooperation must be voluntary: from cooperation the two parties must anticipate payoffs at least as good as their outside options. The remaining conditions reveal two frictions inherent in the exercise of power.

On the one hand, the incumbent faces a *commitment problem*. When the state of the world is good, the incumbent has an incentive to expropriate all the resources, so that, as shown by condition (IC), the incumbent's resource sharing must be dynamically enforced through continuation values that depend on the realized amount of resources received by the opposition. On the other hand, there is *information asymmetry* between the two parties. Because the opposition cannot observe the state, a bad realization is observationally equivalent to the incumbent's expropriation and, as implied by conditions (PK-I) and (PK-O), both events induce the same continuation values.

Proposition 2. *If an equilibrium has sustained cooperation, it has power alternation.*

Proposition 2 closely connects to Kelsen (1929), who argued that the party in power could secure cooperation through concessions without relinquishing power. Our result shows that, under the two frictions—the incumbent's commitment problem and information asymmetry—concessions alone cannot sustain cooperation.

The key idea is that the incumbent's continuation value must decline following unfavorable state realizations, which, in the absence of power alternation, eventually causes cooperation to break down. To see this, suppose, contrary to Proposition 2, that sustained cooperation is achieved in an equilibrium σ without power alternation. In this equilibrium, one party must eventually monopolize power: there exists a $t \geq 0$ such that for all $\tau \geq t$, $a_\tau = 0$ and $I_{\tau+1} = I_\tau = I_t$. Then, conditions (PK-I) and (IC) in Lemma 1, together with $a_t = 0$, imply

$$\mathbb{E}^\sigma\left(U_{I_t}(\sigma|h^{t+1}) \mid h^t, c_{0,t}c_{1,t} = 1, \theta_t = 0\right) = \Gamma_{I_t}^\sigma(0|h^t) \leq \frac{1}{\delta}U_{I_t}(\sigma|h^t) - \frac{1-\delta}{\delta}q < U_{I_t}(\sigma|h^t).$$

Because of the incumbent's commitment problem, disciplining its resource sharing requires sufficiently low continuation values following deviations. Under information asymmetry, such continuation values must also arise on the equilibrium path after bad state realizations.

Because $\theta_t = 0$ with probability $1 - q > 0$, with positive probability, the incumbent's on-path payoff declines at $t + 1$,

$$U_{I_t}(\sigma|h^{t+1}) \leq \frac{1}{\delta}U_{I_t}(\sigma|h^t) - \frac{1-\delta}{\delta} < U_{I_t}(\sigma|h^t).$$

By iterating this argument, for a sufficiently large $\tau > t$, there must be a set of on-path histories at τ that occurs with positive probability following which the incumbent's payoff falls below its outside option,

$$U_{I_t}(\sigma|h^\tau) \leq \frac{1}{\delta^{\tau-t}}U_{I_t}(\sigma|h^t) - \frac{1-\delta^{\tau-t}}{\delta^{\tau-t}}q < \underline{u}_{I_t}.$$

Along such histories, repeated unfavorable state realizations drive the incumbent's payoff from continuing cooperation so low that it would rather deviate to its outside option, contradicting condition (IR-I).

The intuition behind Proposition 2 is reminiscent of Yared (2010) and Li and Matouschek (2013), both of which study environments with information asymmetry in which enforcing promises of concessions entails on-path punishments that ultimately lead to inefficiencies. Broadly, our result is one of the immiseration results typical in dynamic settings with private information (for instance, Thomas and Worrall, 1990; Atkeson and Lucas, 1992; Lipnowski and Ramos, 2020). In Appendix C, we allow the opposition access to additional information captured by a signal generated by the state θ . We show that this proposition still holds whenever this signal fails to perfectly reveals the true realization of the state. Power alternation remains necessary for sustained cooperation as long as the opposition cannot perfectly distinguish a bad state realization from the incumbent's expropriation, so that on-path punishments are inevitable.

3.3. Self-enforcing continuation values

As shown in Lemma 1, the continuation values of the two parties given the amount of resources the opposition receives play important roles in facilitating their cooperation. These continuation values constitute their payoffs from cooperation, indicated by conditions (PK-I) and (PK-O), and discipline the incumbent's resource sharing, reflected by (IC). To answer the questions of whether and how sustained cooperation can be achieved, we must first identify the set of attainable continuation values.

In principle, the two parties can attain a pair of continuation values as long as it is *self-enforcing*: it is derived from a possible equilibrium path in the continuation play. But because of the incumbent's option to transfer power, three cases are possible.

Consider party j 's incumbency and let (γ_j, γ_{1-j}) be a self-enforcing continuation value pair.

First, (γ_j, γ_{1-j}) can be attained without power transfer. In this case, (γ_j, γ_{1-j}) is a payoff pair the two parties receive by the incumbent holding onto power at the end of the current period and both parties then playing a continuation equilibrium starting from party j 's incumbency at the beginning of the next period,

$$(1) \quad (\gamma_j, \gamma_{1-j}) \in \mathcal{E}_j := \{(U_j(\sigma|I_0 = j), U_{1-j}(\sigma|I_0 = j)) : \sigma \in \Sigma\}$$

where $\mathcal{E}_j$ is the set of all possible equilibrium payoff pairs attainable under party j 's incumbency and, therefore, the set of self-enforcing continuation value pairs attainable under party j 's incumbency without power transfer.

Second, (γ_j, γ_{1-j}) can be attained through power transfer. In this case, (γ_j, γ_{1-j}) is a payoff pair the two parties receive by the incumbent transferring power at the end of the current period and both parties then playing a continuation equilibrium starting from party $1 - j$'s incumbency at the beginning of the next period. Two conditions are required here.

On the one hand, the incumbent must be willing to relinquish power, which requires

$$\gamma_j \geq \underline{u}_j.$$

On the other hand, after power transitions, the two parties must receive an equilibrium payoff pair attainable under party $1 - j$'s incumbency, which requires

$$\left(\gamma_{1-j}, \gamma_j + \frac{1-\delta}{\delta} k \right) \in \mathcal{E}_{1-j}.$$

Hence,

$$(2) \quad (\gamma_j, \gamma_{1-j}) \in \mathcal{A}_j := \left\{ (\gamma_j, \gamma_{1-j}) : \gamma_j \geq \underline{u}_j, \left(\gamma_{1-j}, \gamma_j + \frac{1-\delta}{\delta} k \right) \in \mathcal{E}_{1-j} \right\},$$

where $\mathcal{A}_j$ is the set of self-enforcing continuation value pairs attainable under party j 's incumbency through power transfer.

At last, through the public randomization device z , (γ_j, γ_{1-j}) can be attained through randomizing elements in $\mathcal{E}_j$ and $\mathcal{A}_j$.

Therefore, the set of all possible self-enforcing continuation value pairs attainable under

party j 's incumbency must be

$$(3) \quad \bar{\mathcal{E}}_j := \text{co}(\mathcal{E}_j \cup \mathcal{A}_j).$$

Lemma 2. *If $\sigma \in \Sigma$, $h^t \in \mathcal{H}$ is on the equilibrium path of σ , and $C_0^\sigma(h^t)C_1^\sigma(h^t) = 1$, then*

$$(SE) \quad (\Gamma_{I_t}^\sigma(s_t|h^t), \Gamma_{1-I_t}^\sigma(s_t|h^t)) \in \bar{\mathcal{E}}_{I_t}, \forall s_t \in [0, 1].$$

3.4. Necessary and sufficient condition for sustained cooperation

The standard tool for characterizing the set of attainable continuation values in repeated games is the *self-generation approach* pioneered by [Abreu, Pearce and Stacchetti \(1990\)](#). This approach applies in our setting as long as $\bar{\mathcal{E}}_j = \text{co}(\mathcal{E}_j)$, in which case the set of equilibrium payoff pairs under party j 's incumbency generates itself via convexification.

In our setting, however, $\bar{\mathcal{E}}_j = \text{co}(\mathcal{E}_j)$ cannot be taken for granted because of the incumbent's option to transfer power. Evidently in (3), the equality fails as long as $\mathcal{A}_j$ contains at least one element that is not in $\text{co}(\mathcal{E}_j)$. In this case, the two parties can have self-enforcing continuation values via transitioning into party $1 - j$'s incumbency that are not attainable under party j 's incumbency. Consequently, $\bar{\mathcal{E}}_j$ depends both on $\mathcal{E}_j$ and on $\mathcal{E}_{1-j}$ through $\mathcal{A}_j$, implying that $\mathcal{E}_j$ can no longer generate itself.

To address this issue, we extend the self-generation approach of [Abreu, Pearce and Stacchetti \(1990\)](#) to recover a recursive structure, in which the dual of the two sets of equilibrium payoff pairs $(\mathcal{E}_0, \mathcal{E}_1)$ generates itself. This extension allows characterizations of the sets of self-enforcing continuation value pairs corresponding to both incumbencies. Details of this construction are provided in Appendix B.

Lemma 3. *For each $j \in \{0, 1\}$, $\mathcal{E}_j$, $\mathcal{A}_j$, and $\bar{\mathcal{E}}_j$ are compact and, in the order of set inclusion, each of these sets is increasing in δ and decreasing in k , $\underline{u}_j$, and $\underline{u}_{1-j}$.*

Lemma 3 shows the standard properties of the sets of equilibrium payoff pairs, the sets of self-enforcing continuation value pairs attainable through power transfer, and the sets of all possible self-enforcing continuation value pairs. The compactness of $\mathcal{E}_{I_0}$ ensures the existence of at least one efficient equilibrium, $\Sigma^* \neq \emptyset$. The monotonicity of these sets in δ , k , $\underline{u}_0$, and $\underline{u}_1$ help to derive the necessary and sufficient condition for sustained cooperation.

Theorem 1. *There exists an equilibrium with sustained cooperation if and only if $\delta > \delta^*$ and $k \leq k^*$, where $k^* > \underline{k}$ is increasing in δ and decreasing in $\underline{u}_0$ and $\underline{u}_1$. Moreover, if at least one equilibrium with sustained cooperation exists, then every efficient equilibrium has sustained cooperation, power alternation, and is not Schumpeterian.*

Theorem 1 shows that sustained cooperation can be achieved if and only if the two parties are sufficiently patient and the cost of transferring power is not too high. These conditions are easier when the two parties have worse outside options. Moreover, because $k^* > \underline{k}$, sustained cooperation is feasible even when it cannot be attained in any Schumpeterian equilibrium.

A stronger implication of Theorem 1 is that whenever sustained cooperation is feasible, every efficient equilibrium exhibits sustained cooperation, involves power alternation, and is not Schumpeterian. Therefore, all efficient self-enforcing arrangements necessarily combine political concessions with power alternation: parties cooperate in every period, alternate power over time, and make concessions while in power.

The proof of Theorem 1 consists of three parts. First, we adapt the self-generation approach to our model and show that the equilibrium payoff sets $\mathcal{E}_0$ and $\mathcal{E}_1$ can be solved by an iterative algorithm analogous to [Abreu, Pearce and Stacchetti \(1990\)](#). This allows us to prove the monotonicity of $\mathcal{A}_0$ and $\mathcal{A}_1$ in k as in Lemma 3. Second, we establish that every efficient equilibrium has sustained cooperation if and only if power transfer is feasible under both incumbencies: $\mathcal{A}_j \neq \emptyset$ holds for both $j \in \{0, 1\}$. The necessity follows from Proposition 2. For sufficiency, we show that as long as power transfer is feasible under both incumbencies, any equilibrium that fails to attain sustained cooperation is Pareto suboptimal. Finally, we exploit Proposition 1 and the monotonicity of $\mathcal{A}_0$ and $\mathcal{A}_1$ in k to show that these sets are nonempty if and only if $\delta > \delta^*$ and $k \leq k^*$.

Theorem 1, together with its proof, clarifies the roles of political concessions and power alternation in sustaining cooperation. On the one hand, political concessions assure efficiency and facilitate the sustainability of cooperation. By reallocating resources within periods, concessions reduce the frequency of costly power transfers, making the future relationship more valuable to both parties. This is consistent with [Kelsen \(1929\)](#) and the literature on authoritarian power sharing (for instance, [Gandhi and Przeworski, 2007](#); [Svolik, 2012](#); [Powell, 2013](#); [Meng, Paine and Powell, 2023](#); [Egorov and Sonin, 2024](#)): concessions are employed as an efficient option to assure the opposition's cooperation.

On the other hand, under sustained cooperation, the credibility of concessions stems from power alternation: on-path punishments enforcing concessions entail power transitions rather than the breakdown of cooperation. This result unveils a novel rationale for why power alternation occurs despite the alternative of granting concessions: beyond a mechanism of intertemporal redistribution as in Schumpeterian equilibria, it also disciplines concessions while preserving continued cooperation on the equilibrium path.

Crucially, such discipline does not operate through the cost of transferring power. To see this, consider any efficient equilibrium σ . As elaborated in Section 3.2, at any on-path history

h^t , the on-path punishment to enforce $X_{I_t}^\sigma(h^t)$ is through the continuation value $\Gamma_{I_t}^\sigma(0|h^t)$. Now suppose $(\Gamma_{I_t}^\sigma(0|h^t), \Gamma_{1-I_t}^\sigma(0|h^t)) \in \mathcal{A}_{I_t}$, so that the incumbent transfers power after failing to make concessions. In this case, party I_t receives payoff

$$U_{I_t}(\sigma|h^{t+1}) = \Gamma_{I_t}^\sigma(0|h^t) + \frac{1-\delta}{\delta}k$$

at the beginning of period $t+1$ —after losing power and becoming the opposition. This implies that the cost of relinquishing power borne by party I_t in period t is compensated for in the continuation play. As a result, the on-path punishment $\Gamma_{I_t}^\sigma(0|h^t)$ stems from the continuation equilibrium under the new incumbency that begins at period $t+1$.

Therefore, power alternation disciplines concessions under the current incumbency by shifting on-path punishments to the future incumbency. This mechanism contrasts with the models in which the provision of dynamic incentives under information asymmetry entails the use of inefficient punishments on the equilibrium path (for instance, [Green and Porter, 1984](#); [Yared, 2010](#); [Li and Matouschek, 2013](#); [Padró-i Miquel and Yared, 2012](#); [Lipnowski and Ramos, 2020](#)).¹⁶ In particular, the punishments shifted to the future incumbency are endogenously determined by on-path continuation values attained through power transfer. To completely understand how such discipline works, we must characterize how power is exercised and distributed on the path of play in efficient equilibria. In what follows, we provide such a characterization while assuming the conditions of sustained cooperation.

Assumption 2. $\delta > \delta^*$ and $k \leq k^*$.

4. NORMS OF POWER ALTERNATION

We begin by investigating the pattern of power alternation in efficient equilibria: when would power transfer occur and how? These questions can be addressed by geometrically characterizing the sets of self-enforcing continuation values.

¹⁶In our model, the on-path punishments can be provided when the cost k is small. In particular, in Appendix C, we show that there exists $k' \leq k^*$ such that for any $k \leq k'$, every efficient equilibrium attains sequential efficiency on-path in the sense of [Fudenberg, Holmström and Milgrom \(1990\)](#). Furthermore, as $k \rightarrow 0$, every efficient equilibrium is asymptotically first-best.

4.1. Geometry of continuation values

Lemma 4. For each $j \in \{0, 1\}$, $(\gamma_j, \gamma_{1-j}) \in \bar{\mathcal{E}}_j$ holds if and only if $\gamma_j \in [\underline{u}_j, \bar{u}_j]$ and $\gamma_{1-j} \in [\underline{u}_{1-j}, \bar{v}_{1-j}(\gamma_j)]$, where

$$\bar{v}_{1-j}(\gamma_j) := \sup \left\{ \gamma_{1-j} : (\gamma_j, \gamma_{1-j}) \in \bar{\mathcal{E}}_j \right\}$$

is continuous, concave, and satisfies $\bar{v}_{1-j}(\underline{u}_j) > \underline{u}_{1-j}$ and $\bar{v}_{1-j}(\bar{u}_j) = \underline{u}_{1-j}$.

Lemma 4 describes the shape of the set of self-enforcing continuation value pairs corresponding to each incumbency. As a convex and compact subset of $\mathbb{R}^2$, $\bar{\mathcal{E}}_j$ is fully characterized by its frontier, the curve $\bar{v}_{1-j}(u_j)$ when $u_j \in [\underline{u}_j, \bar{u}_j]$, which hereafter is referred to as the *continuation value frontier* of party j 's incumbency.

By definition, $\bar{u}_j$ is the largest possible continuation value party j can have under its own incumbency, while $\bar{v}_{1-j}(\gamma_j)$ is the largest possible continuation value party $1-j$ can have under party j 's incumbency given that party j itself has the continuation value of γ_j . The following result shows the connection between continuation value frontiers and efficient equilibria.

Lemma 5. For any $\sigma \in \Sigma^*$, $h^t \in \mathcal{H}$ on the equilibrium path of σ , and $\theta_t \in \{0, 1\}$,

$$\Gamma_{1-I_t}^\sigma \left(\theta_t X_{I_t}^\sigma(h^t) \mid h^t \right) = \bar{v}_{1-I_t} \left(\Gamma_{I_t}^\sigma \left(\theta_t X_{I_t}^\sigma(h^t) \mid h^t \right) \right).$$

According to Lemma 5, if a continuation value pair is on the equilibrium path of an efficient equilibrium, then as a point in $\mathbb{R}^2$, it must lie on the continuation frontier of the current incumbency. In any efficient equilibrium, consider any period under party j 's incumbency and suppose party j has the continuation value of γ_j following its resource sharing decision. Then, party $1-j$ must have the continuation value of $\bar{v}_{1-j}(\gamma_j)$. There are three possible locations for the point $(\gamma_j, \bar{v}_{1-j}(\gamma_j))$, corresponding respectively to three qualitatively different events regarding power transfer.

First, if $(\gamma_j, \bar{v}_{1-j}(\gamma_j)) \in \text{co}(\mathcal{E}_j)$, then party j would hold onto power at the end of the current period, so that its power persists. Second, if $(\gamma_j, \bar{v}_{1-j}(\gamma_j)) \in \text{co}(\mathcal{A}_j)$, then party j would transfer power with certainty at the end of the current period, so that party $1-j$ becomes the new incumbent in the next period. Third, if $(\gamma_j, \bar{v}_{1-j}(\gamma_j)) \notin \text{co}(\mathcal{E}_j) \cup \text{co}(\mathcal{A}_j)$, then $(\gamma_j, \bar{v}_{1-j}(\gamma_j))$ must be the convex combination of some points in $\mathcal{E}_j$ and some points in $\mathcal{A}_j$. In this case, power transfer must occur with a proper probability at the end of the current period.

Theorem 2. For each $j \in \{0, 1\}$,

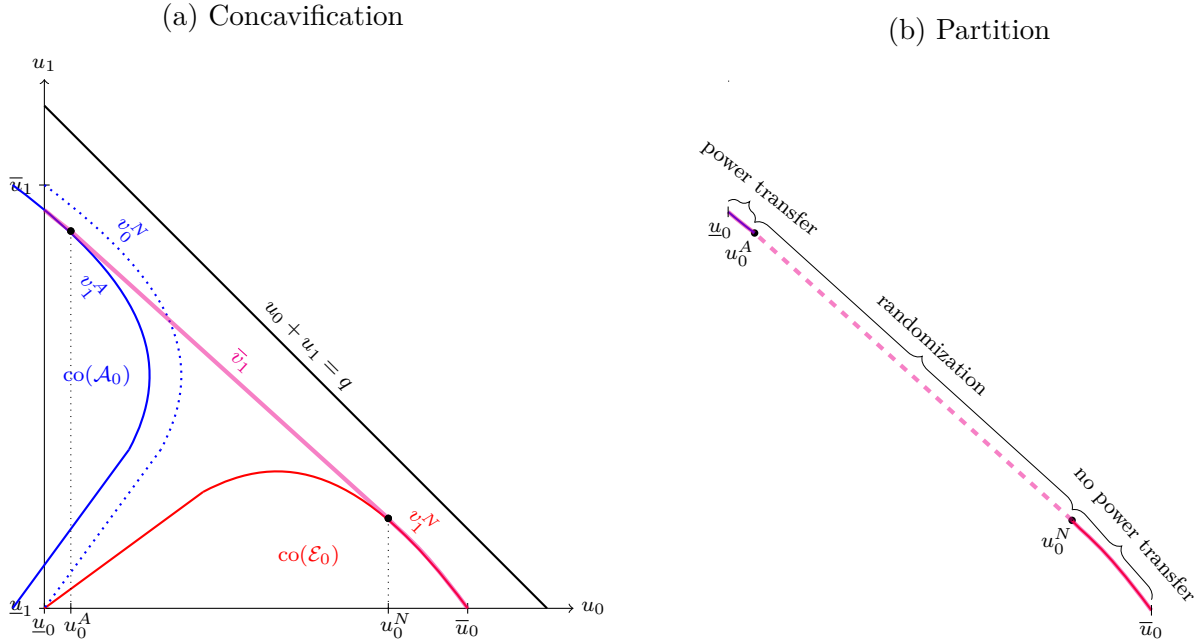
1. $\bar{v}_{1-j}$ is the concavification of $\max\{v_{1-j}^N, v_{1-j}^A\}$, where

$$\begin{aligned} v_{1-j}^N(\gamma_j) &:= \sup \{ \gamma_{1-j} : (\gamma_j, \gamma_{1-j}) \in \text{co}(\mathcal{E}_j) \} \\ v_{1-j}^A(\gamma_j) &:= \sup \{ \gamma_{1-j} : (\gamma_j, \gamma_{1-j}) \in \text{co}(\mathcal{A}_j) \}; \end{aligned}$$

2. there are u_j^N, u_j^A such that $\underline{u}_j \leq u_j^A \leq u_j^N \leq \bar{u}_j$ and

$$\bar{v}_{1-j}(\gamma_j) = \begin{cases} v_{1-j}^N(\gamma_j), & \gamma_j \geq u_j^N \\ \left(1 - \frac{u_j^N - \gamma_j}{u_j^N - u_j^A}\right) v_{1-j}^N(u_j^N) + \frac{u_j^N - \gamma_j}{u_j^N - u_j^A} v_{1-j}^A(u_j^A), & u_j^A < \gamma_j < u_j^N \\ v_{1-j}^A(\gamma_j), & \gamma_j \leq u_j^A. \end{cases}$$

Figure 1: Geometry of continuation values



The above theorem provides a sharp characterization for the continuation value frontiers. By definition, given that party j has the continuation value of γ_j under its own incumbency, $v_{1-j}^N(\gamma_j)$ is the largest continuation value party $1-j$ can have without power transfer, while $v_{1-j}^A(\gamma_j)$ is the largest continuation value party $1-j$ can have through power transfer. Then, as shown by the first part of Theorem 2 and illustrated in Figure 1a, the continuation value frontier of party j 's incumbency, $\bar{v}_{1-j}$, can be obtained by concavifying the joint of v_{1-j}^N and v_{1-j}^A . In other words, $\bar{v}_{1-j}$ is the smallest concave function that everywhere weakly dominates both v_{1-j}^N and v_{1-j}^A .

The second part of Theorem 2 shows that each continuation value frontier can be partitioned into three qualitatively different segments, illustrated in Figure 1b.

Under party j 's incumbency, when it has a sufficiently large continuation value $\gamma_j \geq u_j^N$, $\bar{v}_{1-j}(\gamma_j) = v_{1-j}^N(\gamma_j)$, so that $(\gamma_j, \bar{v}_{1-j}(\gamma_j)) \in \text{co}(\mathcal{E}_j)$. In this case, the pair of continuation values on the frontier, $(\gamma_j, \bar{v}_{1-j}(\gamma_j))$, must be attained with party j holding onto power, choosing $a_t = 0$ at the end of the current period.

When party j has a sufficiently low continuation value $\gamma_j \leq u_j^A$, $\bar{v}_{1-j}(\gamma_j) = v_{1-j}^A(\gamma_j)$, so that $(\gamma_j, \bar{v}_{1-j}(\gamma_j)) \in \text{co}(\mathcal{A}_j)$. In this case, the pair of continuation values on the frontier, $(\gamma_j, \bar{v}_{1-j}(\gamma_j))$, must be attained with party j transferring power, choosing $a_t = 1$ at the end of the current period.

At last, when party j has an intermediate continuation value $\gamma_j \in (u_j^A, u_j^N)$, then $(\gamma_j, \bar{v}_{1-j}(\gamma_j)) \notin \text{co}(\mathcal{E}_j) \cup \text{co}(\mathcal{A}_j)$ lies on the straight line connecting $(u_j^N, v_{1-j}^N(u_j^N))$ and $(u_j^A, v_{1-j}^A(u_j^A))$. In this case, $(\gamma_j, \bar{v}_{1-j}(\gamma_j))$ can be attained by randomizing between $(u_j^N, v_{1-j}^N(u_j^N))$ and $(u_j^A, v_{1-j}^A(u_j^A))$. With probability $(u_j^N - \gamma_j)/(u_j^N - u_j^A)$, the two parties have the continuation values $(u_j^A, v_{1-j}^A(u_j^A))$, which requires party j to transfer power at the end of the current period; with the complementary probability, the two parties have the continuation values $(u_j^N, v_{1-j}^N(u_j^N))$, which requires party j to hold onto power.

4.2. Three norms

The following theorem summarizes our main results regarding the pattern of power alternation in efficient equilibria.

Theorem 3. *For each $j \in \{0, 1\}$,*

1. $u_j^N = \hat{u}_j$, where

$$\hat{u}_j := (1 - \delta)q + \delta \underline{u}_j < \bar{u}_j,$$

so that $(\gamma_j, \bar{v}_{1-j}(\gamma_j))$ is attained without power transfer, $\bar{v}_{1-j}(\gamma_j) = v_{1-j}^N(\gamma_j)$, if and only if $\gamma_j \geq \hat{u}_j$;

2. $u_j^A = \underline{u}_j$, so that $(\gamma_j, \bar{v}_{1-j}(\gamma_j))$ is attained through power transfer, $\bar{v}_{1-j}(\gamma_j) = v_{1-j}^A(\gamma_j)$, if and only if $\gamma_j = \underline{u}_j$.

3. $\gamma_j + \bar{v}_{1-j}(\gamma_j)$ is strictly increasing in γ_j

Norm of backloaded abdication. The first part of Theorem 3 identifies *when* power transfer would and would not occur. Under party j 's incumbency, power transfer would

occur with positive probability when party j 's continuation value falls below $\hat{u}_j$; otherwise, party j would hold onto power. Geometrically, u_j^N in Theorem 2 must coincide with $\hat{u}_j$, so that $\bar{v}_{1-j}(\gamma_j)$ coincides with $v_{1-j}^N(\gamma_j)$ for all $\gamma_j \geq \hat{u}_j$.

By definition, $\hat{u}_j$ is party j 's *minimal cooperative payoff* under its own incumbency: it is the payoff party j secures from expropriating all available resources and then permanently taking its outside option. When party j anticipates a continuation value below $\hat{u}_j$, a positive probability to transfer power is necessary to ensure the continuation of cooperation. If party j holds onto power, then in the next period cooperation must cease with a positive probability, for otherwise party j would get at least $\hat{u}_j$.

Therefore, efficiency implies a *norm of backloaded abdication*, summarized in Corollary 1 below. In any efficient equilibrium, an incumbent retains power whenever possible and relinquishes it only when its continuation value falls so low that power transfer becomes indispensable for sustaining cooperation. In other words, each incumbent prioritizes preserving its power by relying first on political concessions to compensate the opposition and leaving power transfer as a last resort. As a result, incumbencies exhibit persistence, with transfers of power backloaded into the future insofar as cooperation can be sustained without them.

Corollary 1 (Norm of backloaded abdication). *For any $\sigma \in \Sigma^*$, $h^t \in \mathcal{H}$ on the equilibrium path of σ , and $\theta_t \in \{0, 1\}$,*

$$\int_0^1 A_{I_t}^\sigma \left(\theta_t X_{I_t}^\sigma(h^t), z_t \mid h^t \right) dz_t = 0$$

holds if and only if $\Gamma_{I_t}^\sigma \left(\theta_t X_{I_t}^\sigma(h^t) \mid h^t \right) \geq \hat{u}_{I_t}$.

Intuitively, the norm of backloaded abdication arises because resource sharing generates no efficiency loss, whereas power transfer does due to the incumbent's cost of relinquishing power. Relying predominantly on political concessions and delaying power transfer therefore improve efficiency and increase the total surplus generated by cooperation.

Norm of almighty inauguration. The second part of Theorem 3 shows *how* power transfer would occur. If a pair of continuation values on the frontier is attained through power transfer, then the incumbent's continuation value must be equal to its *minimal equilibrium payoff*—its reservation value from the outside option. Geometrically, u_j^A in Theorem 2 must coincide with $\underline{u}_j$, so that $\bar{v}_{1-j}(\gamma_j)$ coincides with $v_{1-j}^A(\gamma_j)$ at a single point $\gamma_j = \underline{u}_j$.

Hence, whenever an incumbent transfers power, it is indifferent between whether to relinquish power and then continue cooperation under the other party's incumbency, or to hold onto power and then permanently take its outside option.

Therefore, efficiency implies a *norm of almighty inauguration*, as summarized in Corollary 2 below. In any efficient equilibrium, the total surplus generated by cooperation under every new incumbency formed after a power transition is divided so as to maximize the share of the new incumbent, with the remainder just compensating the previous incumbent for voluntarily relinquishing power.

Corollary 2 (Norm of almighty inauguration). *For any $\sigma \in \Sigma^*$, $h^t \in \mathcal{H}$ on the equilibrium path of σ , and $\theta_t \in \{0, 1\}$,*

$$\int_0^1 A_{I_t}^\sigma \left(\theta_t X_{I_t}^\sigma(h^t), z_t \mid h^t \right) dz_t = 1$$

holds if and only if $\Gamma_{I_t}^\sigma \left(\theta_t X_{I_t}^\sigma(h^t) \mid h^t \right) = \underline{u}_{I_t}$.

This norm could be interpreted as an implicit bargaining protocol for the continuation equilibrium governing the exercise and distribution of power in the new incumbency. Under such a bargaining protocol, the inaugurating party first makes a take-it-or-leave-it offer for the continuation equilibrium; then the abdicating party chooses whether to accept the offer and subsequently play the proposed continuation equilibrium, or to reject the offer and subsequently receive its reservation value. This protocol renders the inaugurating party the maximal bargaining leverage, making it “almighty.” The inaugurating party would maximize its own payoff in the continuation equilibrium by pushing the abdicating party to indifference.

Recall that the norm of backloaded abdication implies that an incumbent’s continuation value determines its ability to preserve power through making political concessions. The norm of almighty inauguration arises to ensure that every new incumbent, after gaining power, is maximally capable of retaining its power, thereby delaying the subsequent transfer of power and improving efficiency.

Norm of minimal compromise. The third part of Theorem 3 shows how the total surplus generated by the continuation cooperation depends on the incumbent’s continuation value, which aggregates and inversely measures the extent of its compromise to the opposition. For any pair of continuation values on the frontier, the sum of the two values is strictly increasing in the incumbent’s continuation value. Any compromise that lowers the incumbent’s continuation value therefore reduces the total surplus from the continuation cooperation. Geometrically, this means that the slope of the frontier, the curve $\bar{v}_{1-j}$, is everywhere strictly flatter than -1 .

Hence, maximizing the total surplus from the continuation cooperation is equivalent to maximizing the incumbent’s continuation value. Efficiency therefore requires a *norm of minimal compromise*, summarized in Corollary 3 below: incumbents should always

compromise minimally with their oppositions. In particular, among all equilibria that yield the incumbent the same current payoff, an efficient equilibrium maximizes its continuation value following a bad state. Conditional on this, among all equilibria that also deliver the same continuation value following a bad state, an efficient equilibrium further maximizes the incumbent's continuation value following a good state. This norm complements the norm of almighty inauguration by showing that, in any efficient equilibrium, the total surplus generated by cooperation in any continuation game within an incumbency is divided so as to maximize the share of the incumbent.

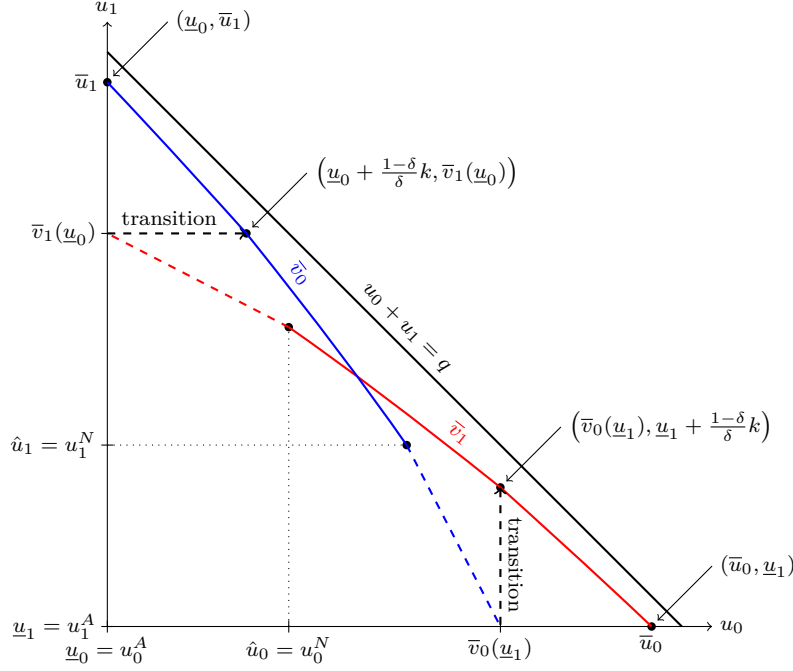
Corollary 3 (Norm of minimal compromise). *For any $\sigma \in \Sigma^*$ and h^t on the equilibrium path of σ , then for any $\sigma' \in \Sigma$ such that $C_0^{\sigma'}(h^t)C_1^{\sigma'}(h^t) = 1$:*

1. $\Gamma_{I_t}^\sigma(0|h^t) \geq \Gamma_{I_t}^{\sigma'}(0|h^t)$ if $U_{I_t}(\sigma'|h^t) = U_{I_t}(\sigma|h^t)$;
2. $\Gamma_{I_t}^\sigma(X_{I_t}^\sigma(h^t)|h^t) \geq \Gamma_{I_t}^{\sigma'}(X_{I_t}^{\sigma'}(h^t)|h^t)$ if $U_{I_t}(\sigma'|h^t) = U_{I_t}(\sigma|h^t)$ and $\Gamma_{I_t}^{\sigma'}(0|h^t) = \Gamma_{I_t}^\sigma(0|h^t)$.

Similarly with the necessity of power alternation for sustained cooperation, the norm of minimal compromise arises from the incumbent's commitment problem and information asymmetry—the two frictions inherent in the exercise of power. Intuitively, a more generous compromise, represented by a lower continuation value of the incumbent, consists of a more generous promise of resource sharing in the next period or a larger chance of power transfer at the end of the current period. Power transfer directly generates an efficiency loss. Resource sharing itself generates no efficiency loss, but enforcing such promises under information asymmetry requires the incumbent to be punished with lower continuation values after unfavorable realizations of the state. By the norm of backloaded abdication, this ultimately leads to a power transfer once the incumbent's continuation value becomes sufficiently low.

In sum, the three norms in Theorem 3 pin down the exact locations of u_j^N and u_j^A and the shape of $\bar{v}_{1-j}$ for both $j \in \{0, 1\}$ and, therefore, allow a proper visualization of the two continuation value frontiers, as shown in Figure 2. More importantly, these norms together characterize how power is exercised and distributed in efficient self-enforcing arrangements. The norm of backloaded abdication governs when power is relinquished: an incumbent transfers power only when its continuation value falls below the level required to sustain cooperation under its incumbency. The norm of almighty inauguration implies that each new incumbent begins with the maximal continuation value, thereby maximizing its capacity to preserve its incumbency through future concessions. Finally, the norm of minimal compromise implies that within each incumbency the incumbent exercises power so as to maximize its continuation value and thus its ability to retain power. Taken

Figure 2: Continuation value frontiers



together, these norms imply that efficient self-enforcing arrangements maximize the stability of power, with alternation occurring only when necessary to sustain cooperation.

As we have elaborated in Theorem 1, power alternation disciplines concessions through continuation values under future incumbencies. Theorem 3 and its corollaries above further elucidate how such discipline works. Under each incumbency, transfer of power is left as a last resort, and once it occurs, on-path punishments shifted to the future incumbency are determined by an implicit protocol for the continuation play that endows the inaugurating party with the maximal bargaining leverage. As a result, the abdicating incumbent expects a continuation value that compensates just enough for its relinquishing power and continuing cooperation. Such punishments are delivered by the new incumbent who will compromise minimally and maximize its own ability to retain power throughout its new incumbency. The stability of power thus arises from the discipline of political concessions in efficient self-enforcing arrangements.

5. DYNAMICS IN EFFICIENT EQUILIBRIA

The three norms of power alternation are informative about the equilibrium payoff pair the two parties receive at the beginning of every period in an efficient equilibrium.

Corollary 4. *For any $\sigma \in \Sigma^*$ and $h^t \in \mathcal{H}$ on the equilibrium path of σ ,*

1. $U_{I_t}(\sigma|h^t) \in [\hat{u}_{I_t}, \bar{u}_{I_t}]$ and $U_{1-I_t}(\sigma|h^t) = \bar{v}_{1-I_t}(U_{I_t}(\sigma|h^t))$;
2. if $t = 0$, then $U_{I_0}(\sigma|h^0) = \bar{u}_{I_0}$ and $U_{1-I_0}(\sigma|h^0) = \underline{u}_{1-I_0}$;
3. if $I_{t-1} \neq I_t$, then $U_{I_t}(\sigma|h^t) = \bar{v}_{I_t}(\underline{u}_{1-I_t}) < \bar{u}_{I_t}$ and $U_{1-I_t}(\sigma|h^t) = \underline{u}_{1-I_t} + (1 - \delta)/\delta k$.

According to the first part of Corollary 4, the equilibrium payoff pair in any efficient equilibrium always lies on the continuation value frontier of the current incumbency with the incumbent's payoff varying between $\hat{u}_{I_t}$ and $\bar{u}_{I_t}$. Hence, the identity and payoff of the incumbent together pin down the location of the equilibrium payoff pair. As illustrated in Figure 2, the equilibrium payoff pair moves along the solid red curve under party 0's incumbency and along the solid blue curve under party 1's incumbency.

The second part of Corollary 4 identifies the equilibrium payoff pair in the initial period: the initial incumbent receives its largest possible equilibrium payoff $\bar{u}_{I_0}$, while the initial opposition receives its reservation value $\underline{u}_{I_0}$. Hence, in Figure 2, the dynamics begin either at $(\bar{u}_0, \underline{u}_1)$ when $I_0 = 0$ or at $(\underline{u}_0, \bar{u}_1)$ when $I_0 = 1$. This result follows directly from the norm of minimal compromise. The initial incumbent compromises minimally with the initial opposition and obtains its maximal equilibrium payoff by making the opposition indifferent between cooperation and its outside option.

The third part of Corollary 4 identifies the equilibrium payoff pair in any period immediately after a power transition: the new incumbent receives a payoff $\bar{v}_{I_t}(\underline{u}_{1-I_t}) < \bar{u}_{I_t}$ strictly lower than its maximal equilibrium payoff, while the new opposition receives a payoff $\underline{u}_{1-I_t} + (1 - \delta)/\delta k > \underline{u}_{1-I_t}$ strictly larger than its reservation value. Hence, in Figure 2, the dynamics of every new incumbency of party 0 begin at the point $(\bar{v}_0(\underline{u}_1), \underline{u}_1 + (1 - \delta)/\delta k)$ and that of every new incumbency of party 1 begin at $(\underline{u}_0 + (1 - \delta)/\delta k, \bar{v}_1(\underline{u}_0))$. This result reflects the norm of almighty inauguration. The inaugurating party maximizes its equilibrium payoff at the beginning of the new incumbency under the constraint of ensuring the abdicating party at least its reservation value at the moment of transferring power.

Comparing the last two results yields an interesting observation: a new incumbent after a power transition gets an equilibrium payoff strictly smaller than what the same party could get as the initial incumbent. This is because power transfer is costly and voluntary. Hence, the new incumbent must compensate the previous incumbent not only for cooperation but also for relinquishing power. Indeed, the new opposition gets an equilibrium payoff equal to its reservation value plus $(1 - \delta)/\delta k$, which exactly compensates its cost to relinquish power in the last period.

In the remainder of this section, we first show that in any efficient equilibrium, the current incumbent's identity and its equilibrium payoff at the beginning of each period together

constitute a sufficient statistic for the history and fully determines the behaviors of the two parties in the current period and their expectations of the incumbent's equilibrium payoff at the beginning of the next period. Then, we distinguish three types of political dynamics that differ in the stability of incumbencies and could endogenously arise in efficient equilibria.

5.1. Political capital and behaviors

Proposition 3. *For any $\sigma \in \Sigma^*$, $j \in \{0, 1\}$, and $h^t \in \mathcal{H}$ on the equilibrium path of σ such that $I_t = j$, let $u_j = U_j(\sigma|h^t) \in [\hat{u}_j, \bar{u}_j]$, then*

1. *if $\theta_t = 0$, then the incumbent anticipates the continuation value of*

$$\Gamma_j^\sigma(0|h^t) = \psi(u_j) := \frac{1}{\delta}u_j - \frac{1-\delta}{\delta}q < u_j$$

and transfers power with probability

$$\int_0^1 A_j^\sigma(0, z_t|h^t) dz_t = \alpha_j(u_j) := \max \left\{ \frac{\hat{u}_j - \psi(u_j)}{\hat{u}_j - \underline{u}_j}, 0 \right\},$$

which is decreasing in u_j and is positive if and only if

$$u_j < u_j^\dagger := (1 - \delta^2)q + \delta^2 \underline{u}_j;$$

2. *if $\theta_t = 1$, then the incumbent shares the amount of resources*

$$X_j^\sigma(h^t) = \chi_j(u_j) := \min \left\{ \frac{\delta}{1-\delta} (\bar{u}_j - \psi(u_j)), 1 \right\},$$

which is decreasing in u_j and is always positive, then the incumbent anticipates the continuation value of

$$\Gamma_j^\sigma(X_j^\sigma(h^t) | h^t) = \psi(u_j) + \frac{1-\delta}{\delta} \chi_j(u_j) \geq u_j$$

and holds onto power,

$$\int_0^1 A_j^\sigma(X_j^\sigma(h^t), z_t | h^t) dz_t = 0.$$

By Proposition 3, in any efficient equilibrium σ , the decisions of the two parties at every t and their continuation values depend on the history h^t only through a *state vector* (I_t, u_{I_t}) , where $I_t \in \{0, 1\}$ is the identity of the current incumbent and $u_{I_t} = U_{I_t}(\sigma|h^t) \in [\hat{u}_{I_t}, \bar{u}_{I_t}]$ is

the incumbent's equilibrium payoff at the beginning of t . Hence, in Figure 2, the dynamics in σ can be tracked by the trajectory of (I_t, u_{I_t}) , where I_t indicates the curve, red for $I_t = 0$ and blue for $I_t = 1$, and u_{I_t} identifies the location, the x-coordinate on the red curve and the y-coordinate on the blue curve.

The incumbent's equilibrium payoff u_{I_t} encodes the level of trust it is credited by the opposition depending on the history of their interactions prior to t . We thus refer to u_{I_t} as the incumbent's “*political capital*” at t .¹⁷

According to Proposition 3, the incumbent's political capital first determines the amount of resources it shares with the opposition when the state of the world is good, $\chi_{I_t}(u_{I_t})$. Because $\chi_{I_t}(u_{I_t}) > 0$, resource sharing is always employed. Because $\chi_{I_t}(u_{I_t})$ is decreasing in u_{I_t} , a higher level of political capital allows the incumbent to share less. Interestingly, when the incumbent has a sufficiently low level of political capital,

$$(4) \quad u_{I_t} \leq \delta \bar{u}_{I_t} - (1 - \delta)(1 - q),$$

$\chi_{I_t}(u_{I_t}) = 1$, so that the opposition would receive all available resources.

Second, if the state is favorable, $\theta_t = 1$, and, therefore, the opposition receives the resources of $s_t = \chi_{I_t}(u_{I_t})$, the incumbent holds onto power and, at the beginning of the next period, it expects an increase of its political capital to

$$\psi(u_{I_t}) + \frac{1 - \delta}{\delta} \chi_{I_t}(u_{I_t}) \geq u_{I_t}.$$

When the incumbent has a sufficiently low level of political capital for which (4) holds, it is willing to share all available resources with the opposition because this allows the incumbent to improve its political capital, so it can share less with the opposition if the state is good in the next period. When the incumbent instead has a sufficiently high level of political capital such that (4) fails, it shares an amount, $\chi_{I_t}(u_{I_t}) < 1$, which by definition is just enough to increase its political capital in the next period to its maximal level,

$$\psi(u_{I_t}) + \frac{1 - \delta}{\delta} \chi_{I_t}(u_{I_t}) = \bar{u}_{I_t}.$$

Third, in the case the state is unfavorable, $\theta_t = 0$, and, therefore, the opposition receives

¹⁷The dynamics of political capital are reminiscent of Dixit, Grossman and Gul (2000). Similar to our model, the strength of an incumbent's power also evolves over time, determining the prospect of power switches and, consequently, concessions that the incumbent can credibly make. However, the strength of political power is a state of their stochastic game and its transition follows an exogenous Markov process. In contrast, in our model, the formation and evolution of political capital is an equilibrium phenomenon; it is interpretive, rather than a part of the game.

no resources, $s_t = 0$, the incumbent's political capital determines its probability to transfer power, $\alpha_{I_t}(u_{I_t})$. When the incumbent does transfer power, $a_t = 1$, a new incumbency starts in the next period and party $1 - I_t$ begins with the political capital of $\bar{v}_{1-I_t}(\underline{u}_{I_t})$. When the incumbent holds onto power, $a_t = 0$, it expects a decrease of its political capital to

$$\psi(u_{I_t}) < u_{I_t},$$

for which it must share more resources with the opposition if the state of the world turns out to be good in the next period.

Most importantly, the incumbent has a positive probability to transfer power at the end of t only if it has a sufficiently low level of political capital at the beginning of t , $u_{I_t} < u_{I_t}^\dagger$, and fails to transfer resources to the opposition, $s_t = 0$. In other words, as long as the incumbent has enough political capital, $u_{I_t} \geq u_{I_t}^\dagger$, it faces no risk of losing power regardless of the realization of the state.

As shown in Proposition 3, on the equilibrium path of any efficient equilibrium, the continuation value the incumbent receives following its resource sharing decision is at least $\psi(u_{I_t})$. By definition, $u_{I_t} \geq u_{I_t}^\dagger$ implies $\psi(u_{I_t}) \geq \hat{u}_{I_t}$, so that in the next period the two parties are still able to cooperate under party I_t 's incumbency. The norm of backloaded abdication therefore requires the incumbent to hold onto power.

In contrast, $u_{I_t} < u_{I_t}^\dagger$ implies $\psi(u_{I_t}) < \hat{u}_{I_t}$, so that after a bad realization of the state, the incumbent's continuation value falls below $\hat{u}_{I_t}$. In this case, ensuring the continuation of cooperation requires the incumbent to have a positive probability of transferring power.

To summarize, recall that the incumbent has two ways to compensate for the opposition's voluntary cooperation: political concessions in the form of resource sharing and power transfer. According to Proposition 3, both are employed in each efficient equilibrium. In particular, the incumbent promises different levels of political concessions depending on its political capital, expecting the political capital to accumulate whenever the promise is kept and deplete whenever the promise is broken. The decrease in political capital, in turn, results in more generous promises of concessions. The incumbent keeps relying only on concessions to compensate the opposition until consecutively breaching its promises, after which its political capital falls too low, rendering power transfer indispensable for the continuation of cooperation.

5.2. Three types of political dynamics

Proposition 3 shows that the incumbent suffers no risk of losing power at any period it enjoys a sufficiently high level of political capital, $u_{I_t} \geq u_{I_t}^\dagger$. In this case, we referred to

the incumbency as “*hegemonic*.” In contrast, with a sufficiently low level of political capital, $u_{I_t} < u_t^\dagger$, the incumbent must bear with a positive probability of losing power after a bad realization of the state. In this case, we referred to the incumbency as “*precarious*.” Based on these findings, we can further elaborate on the political dynamics in efficient equilibria under the assumption of symmetry.

Assumption 3. $\underline{u}_0 = \underline{u}_1 = \underline{u}$.

With symmetry, the subscripts that indicate party identities can be omitted. Moreover, we show that depending on two criteria, three types of political dynamics that differ in the stability of incumbencies could endogenously emerge in efficient equilibria.

The first criterion concerns the existence of hegemonic incumbency. The maximal level of political capital each incumbent can achieve is $\bar{u}$. Hence, hegemonic incumbency is attainable if and only if

$$(5) \quad \bar{u} \geq u^\dagger.$$

The second criterion concerns whether each new incumbency following a power transition begins as hegemonic. According to Corollary 2, following a power transition, a new incumbent begins with the political capital of $\bar{v}(\underline{u})$. Hence, each new incumbency is hegemonic if and only if

$$(6) \quad \bar{v}(\underline{u}) \geq u^\dagger.$$

Because $\bar{v}(\underline{u}) < \bar{u}$, (6) implies (5), so that three cases are possible.

Unstable political dynamic. If (5) fails, hegemonic incumbency is unattainable.

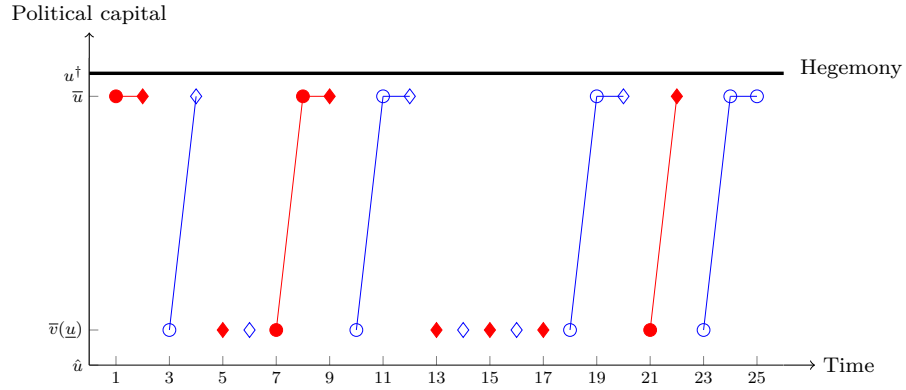
In this case, every incumbency is precarious, so that power transition could happen in each period as it would occur with positive probability following a bad state. The resulting power dynamic, therefore, is *unstable*, featuring short incumbencies and frequent power transitions. This type of political dynamic is illustrated in Figure 3a.

Stable political dynamic. If (6) holds, not only is hegemonic incumbency attainable, every new incumbency begins as hegemonic.

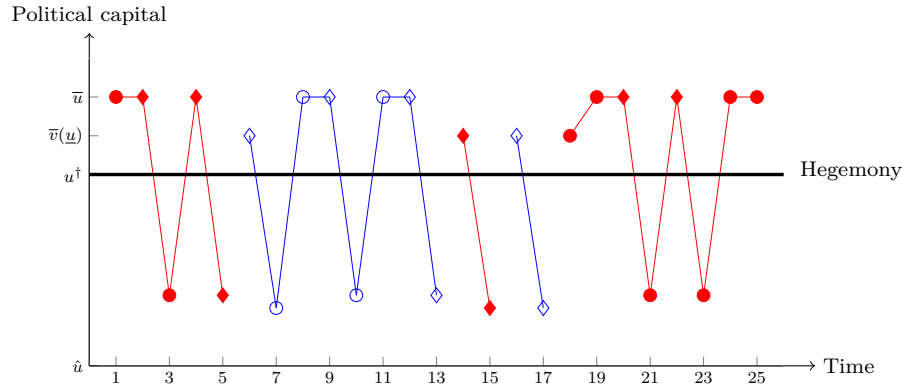
In this case, power transition never happens consecutively: the incumbency in any period following a power transition is hegemonic, so that power transition would not occur in that period regardless of the state. The resulting political dynamic is *stable*, which, to the opposite of unstable political dynamic, features long incumbencies and infrequent power transitions. This type of political dynamic is illustrated in Figure 3b.

Figure 3: Three types of political dynamics

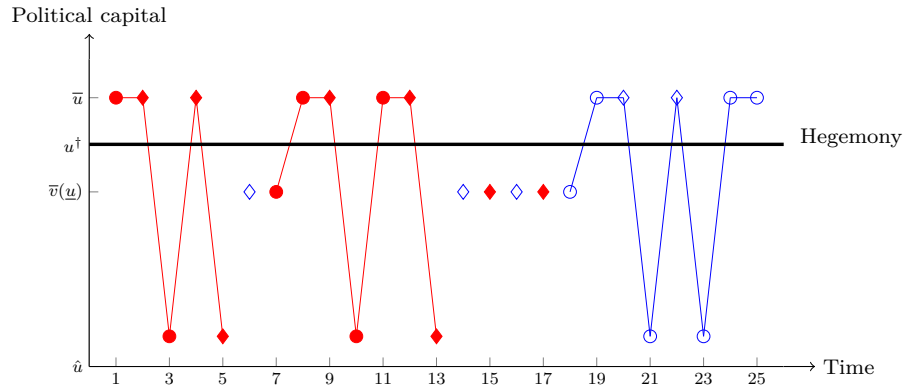
(a) Unstable



(b) Stable



(c) Stabilizable



Notes: Dots represent periods under good states, while diamonds for periods under bad states. Different colors correspond to different incumbents. Points connected by solid lines belong to the same incumbency. All three figures are based on an identical sequence of state realizations, allowing for a controlled comparison across different types of political dynamics.

Stabilizable political dynamic. The third and most interesting case is when (5) holds but (6) fails.

In this case, each new incumbency following a power transition must begin as precarious, but has the potential to stabilize into a hegemony. This political dynamic, therefore, is referred to as *stabilizable*.

Under this type of political dynamic, new incumbencies are vulnerable: a single unfavorable realization of the state could force any new incumbent to transfer its newly acquired power. Nevertheless, if a new incumbent is fortunate enough to experience consecutive good states, it would gradually build up a hegemony, after which it suffers no immediate risk of losing power. As a result, the stabilizable political dynamic, as illustrated in Figure 3c, features cycles between “stable phases” in which one party successfully establishes a hegemony and thus firmly holds onto power and “unstable phases” in which neither party succeeds in building a hegemony and thus power alternates frequently.

The stabilizable political dynamic could arise because the transfer of power is costly and must be voluntary. Even if hegemony is attainable at the maximal level of political capital, ensuring the willingness of the previous incumbent to relinquish power requires each new incumbent to begin the new incumbency with a compromised level of political capital, which could be too low for hegemony.

Proposition 4. *If*

$$\underline{u} \geq \underline{u}^H := \frac{\delta^2 q}{1 + \delta^2}$$

then, for all $k \leq k^$, $\bar{u} < u^\dagger$ and the political dynamic is unstable in efficient equilibria. Moreover, if $\underline{u} < \underline{u}^H$, then there are k^L, k^H such that $0 < k^L \leq k^H \leq k^*$ and*

- 1. if $k \leq k^L$, then $\bar{v}(\underline{u}) \geq u^\dagger$ and the political dynamic is stable in efficient equilibria;*
- 2. if $k^L < k \leq k^H$, then $\bar{v}(\underline{u}) < u^\dagger \leq \bar{u}$ and the political dynamic is stabilizable in efficient equilibria;*
- 3. if $k^H < k \leq k^*$, then $\bar{u} < u^\dagger$ and the political dynamic is unstable in efficient equilibria.*

Furthermore, $k^L < k^H < k^$ if $\underline{u} \in (\underline{u}^L, \underline{u}^H)$, where $\underline{u}^L \in (0, \underline{u}^H)$.*

Proposition 4 shows that which type of political dynamic would arise in efficient equilibria is jointly determined by the reservation value the two parties receive from their outside options, $\underline{u}$, and the cost of power transfer k .

On the one hand, the maximal level of political capital each incumbent is able to achieve, $\bar{u}$, matters for whether the political dynamic is unstable. This level of political

capital, as shown in Corollary 3, requires a compromise that is just enough to compensate for the opposition's voluntary cooperation and, therefore, depends on how much the opposition expects to gain from the future cooperation and its outside option. An extreme implication of Proposition 4 is that when the outside options are sufficiently good, hegemony is unattainable, resulting in an unstable political dynamic, regardless of the cost of power transfer. In fact, condition $\underline{u} \geq \underline{u}^H$ ensures that $\bar{u} < u^\dagger$ holds even when power transfer is costless.¹⁸

On the other hand, the level of political capital of each new incumbent following a power transition, $\bar{v}(\underline{u})$, matters for whether the political dynamic is stable or stabilizable. This level of political capital, as shown in Corollary 4, requires a larger compromise that is sufficient to additionally compensate for the predecessor to voluntarily relinquish power and, therefore, depends on the cost of power transfer. As shown in Proposition 4, so long as $\underline{u} < \underline{u}^H$, the cost of power transfer matters for the type of political dynamic.

First, when the cost of power transfer is extremely low, $k \leq k^L$, the political dynamic is stable. In this case, voluntary cooperation does not require too large a compromise to compensate because of the unattractive outside options, while voluntary power transfer does not require much more because of its extremely low cost. As a result, each new incumbent is able to begin with a level of political capital sufficiently high for a hegemony.

Second, when the cost of power transfer is intermediate, $k^L < k \leq k^H$, the political dynamic is stabilizable. Here, the unattractive outside options enable each incumbent to easily compensate the opposition's voluntary cooperation, rendering hegemony attainable. The relatively high cost of power transfer, however, implies that each new incumbent must make a substantially larger compromise to compensate for the previous incumbent's voluntary power transfer, for which it can only begin with a level of political capital insufficient for a hegemony.

At last, when the cost of power transfer is very high, $k > k^H$, the political dynamic is again unstable. In this case, each time power transition occurs, a very large compromise must be made for the previous incumbent to be willing to burden the very high cost to relinquish power, which results in a very low level of political capital for the new incumbent. With limited political capital, the new incumbent could quickly lose power and, for its voluntary decision to relinquish power, the next incumbent must again make a very large compromise, and so on. According to the norm of minimal compromise, these compromises generate substantial efficiency losses. As a result, the total surplus the two parties gain from sustained cooperation is limited, so as the maximal level of political capital each incumbent is able to accumulate.

¹⁸When $k = 0$, $\bar{u} = q - \underline{u}$, so that $\bar{u} \leq u^\dagger$ is equivalent to $\underline{u} \geq \underline{u}^H$.

Hence, Proposition 3 implies that the stability of power deteriorates as the cost of power transfer increases. This result clarifies the mechanism for power alternation to discipline concessions: as emphasized in Theorem 1, it operates through shifting on-path punishments to relationships under future incumbencies, rather than imposing the cost of relinquishing power on incumbents. It stands in contrast with papers in which on-path punishments under information asymmetry are explicitly provided through the use of costly instruments—for instance, war in Yared (2010) and direct control in Padró-i Miquel and Yared (2012). In these papers, as the inefficient punishments become more severe, the sustainability of cooperative outcomes expands and the instruments are used less frequently on the equilibrium path. However, in our model, transfer of power is voluntary and must be compensated for after the transition. As a result, as power transfer becomes more costly, every inaugurating incumbent has to make larger compromises to its predecessor, which undermines its own ability to retain power, thereby leading to more frequent alternation and reducing the total surplus from sustained cooperation.

6. HISTORICAL CONTEXTS

This section briefly reviews historical evidence of power alternation and shows how it aligns with the model’s core assumptions, mechanisms, and equilibrium patterns. The early stages of democratization in Britain and Sweden fit this framework well: stable alternation first arose from elite bargains, and only later did these arrangements extend to a broader electorate.

6.1. *Britain, 1688-1832*

Britain provides an early laboratory for observing how elite factional strife coalesced into party-based alternation in power. The Whig–Tory divisions grew out of seventeenth-century political, religious, and economic conflicts that gave rise to the Glorious Revolution of 1688 (Harris, 2006; Pincus, 2009). The constitutional settlement created a favorable setting for the expansion of parliamentary politics, though electoral participation remained highly restricted to the elite stratum (Plumb, 1967).

Between 1688 and 1832, Britain navigated a volatile geopolitical environment shaped by war and imperial expansion. It rose rapidly to global prominence—culminating in victory during the Seven Years’ War—only to suffer the loss of the American colonies soon after (Colley, 1992; Simms, 2008). Party identities gradually took shape against this backdrop. The Whigs advocated an active role in continental warfare, whereas the Tories were more skeptical of continental engagement and placed greater emphasis on colonial interests

(Colley, 1982). The ruling party, entrusted with policymaking, warfare, and diplomacy, often operated behind a veil of secrecy justified on grounds of national security. As Simms (2008) notes, when critics in Parliament called for a select committee to inspect official papers, the Walpole administration replied:

“Parliament ought in no case to call for those papers which may contain secrets, the publishing whereof might be of signal disadvantage to the commonwealth.¹⁹ The danger was that publication of secret documents might perhaps open new sores and give offence to some powers with whom we have at present a good understanding.”

This exacerbated information asymmetries and set off a cycle of mutual suspicion. The Tories often accused the Whigs of waging unnecessary wars and exploiting financial schemes for private gain, while the Whigs charged the Tories with conducting secret and corrupt negotiations with rival powers to end wars at the expense of national interests (Colley, 1982). At the same time, rapid territorial and commercial expansion vastly increased the ruling party’s control over distributive power (Brewer, 1989):

“Growing wealth, frequent wars and the acquisition of an empire had created an enormous executive, and thereby enormous powers of patronage for the government. Sectarian and particular interests—notably those of financiers, merchant contractors and stock-jobbers—were tied to administration. Similarly placemen—Treasury officials, members of the diplomatic and colonial services—as well as pensioners, courtiers and members of the armed forces were dependent for at least part of their livelihood upon the good offices. . . .”

The opposition’s leverage lay in its capacity to withhold trust—a pattern evident in opposition pamphlets, parliamentary criticisms, and votes of no confidence (Brewer, 1976). Over time, a ministry’s survival came to depend less on royal favor and more on maintaining parliamentary confidence, reinforcing the emerging convention of ministerial responsibility (Langford, 1989). Ministries often resigned after repeatedly failing to defend their policies in the wake of major military or political setbacks. The Duke of Marlborough fell amid disputes over the War of the Spanish Succession; Robert Walpole after the unpopular War of Jenkins’ Ear; Lord Carteret over the burdens of the War of the Austrian Succession; the Duke of Newcastle after the loss of Minorca in 1756; and William Pitt the Elder for urging confrontation with Spain during the Seven Years’ War (O’Gorman, 1997; Simms, 2008).

¹⁹Walpole’s use of “commonwealth” shows that contemporary elites recognized their shared interests.

Power turnover was not without its costs. British politics was most stable during periods of one-party hegemony—Whig ascendancy (1714–1760) and Pittite–Tory dominance (mid-1780s–1820)—and most volatile in the two decades after George III’s accession (1760–1782). As [Simms \(2008\)](#) observes, this frequent turnover took a toll:

“The constant changes of ministry, and especially of secretaries of state for foreign affairs, was extremely disruptive. It contrasted sharply with the stability of the early eighteenth century. . . . European policy was in the hands of a succession of young and inexperienced secretaries of state. . . . ”

Accordingly, norms regulating alternation emerged through pragmatic adaptation rather than institutional design.²⁰ The first norm—the “norm of almighty inauguration”—arose when incoming ministers increasingly insisted on greater control over cabinet formation upon taking office. It began as ministers like Walpole “exerted significant influence” in forming ministries “more closely tied to party fortunes” so they could “secure working majorities in the Commons” ([Hoppit, 2002](#)). In contrast, earlier monarchs wanted “mixed ministries.” [Brewer \(1976\)](#) describes how William Pitt the Elder assumed office “only if he was given *carte blanche* (complete freedom to act). . . . [Given] total control over the new administration, he was *generalissimo* (supreme commander) once more.” Another example comes from the Rockingham Whigs, who insisted on party control over cabinet:

“Rockingham’s conduct during the abortive negotiation of 1780, and again at the foundation of the Rockingham administration of 1782, was dominated by the desire to ensure party hegemony within any new ministry. . . . [If not] a clean sweep to force. . . . [it preferred] refusal to come into power.”

The Rockingham Whigs are often seen as marking the emergence of a coherent party ideology and the norm of party-based cabinet formation ([Brewer, 1976](#); [O’Gorman, 1975](#); [Skjönsberg, 2021](#)). As [Langford \(1973\)](#) notes, “in the second Rockingham Ministry Fox’s doctrine [held] that the party in power should in effect choose its own leader and direct both measures and men.” By the nineteenth century, electoral reforms further strengthened party discipline and reinforced its decisive role in cabinet selection ([Briggs, 1959](#)).

Meanwhile, incumbent elite factions routinely resorted to the second norm—the “norm of backloaded abdication”—to delay stepping down in the face of adverse shocks. Such

²⁰This was a period in which parties coexisted with, and gradually evolved out of, elite factions. Our theory accommodates situations in which a single party rises to hegemony for multiple periods; in such cases, alternation can naturally occur between factions within the dominant party. The norms thus apply both to alternation between parties and to alternation between factions within a party.

tactics ranged from mid-term concessions on policy or office to efforts to buy off, placate, or co-opt their opponents. As [Simms \(2008\)](#) shows, the Newcastle ministry, confronted with the political fallout from the loss of Minorca, yielded to opposition demands for inquiry and scapegoating:

“Newcastle’s parliamentary position began to buckle under the failure of his foreign policy. . . there was a steady haemorrhaging of confidence in his ministry. The government was forced, with some misgivings, to establish an inquiry into the loss of Minorca and to prosecute Byng. . . Byng was duly found guilty, and shot on his own quarterdeck in March of the following year.”

The succession of military failures nevertheless fatally weakened Newcastle’s ministry, making it clear by year’s end that he would be forced to step aside. He had little choice but to accept his long-time opponent, William Pitt the Elder, into the cabinet, since “reconciliation with Pitt was the only way to regain office.” As Newcastle himself put it:

“I think Pitt must come. He will come as a conqueror. I always dreaded it. But I had rather be conquered by an enemy who can do our business.”

[Skjönsberg \(2021\)](#) also observes that Pitt “would likely have remained in opposition” had it not been for two major foreign-policy shocks—the threat of invasion and the loss of Minorca in June 1756. Comparable cross-party arrangements emerged in cases such as the Fox–North coalition (1783) and Canning’s coalition (1827), each formed in the aftermath of a fallen ministry. We likewise find instances of cooptation, in which “opposition members were lured with places and pensions” ([Skjönsberg, 2021](#)), as well as evidence of policy concessions, including the Harley ministry’s being pressured to pass the Land Qualification Bill ([Plumb, 1967](#)) and the Walpole ministry’s abandonment of the proposed Excise Scheme ([Plumb, 1956a](#)).

6.2. Sweden, 1719-1772

As [Roberts \(2003\)](#) shows, eighteenth-century Sweden developed one of Europe’s most distinctive early experiments in parliamentary government—an arrangement that, in several respects, advanced beyond Britain’s post-1688 settlement. The constitutional reforms of 1719–20 transferred effective authority to the Riksdag and its committees, where political power rested largely with the noble estate and senior administrators, while the Crown retained only limited initiative. Ministers answered directly to Riksdag scrutiny, particularly through the Secret Committee, giving Sweden an unusually early and rigorous

form of parliamentary accountability. Within this framework, political life cohered around two elite factions: the Caps, associated with peace, fiscal restraint, and cautious diplomacy; and the Hats, who promoted commercial expansion, economic activism, and a more assertive international posture. Their successive periods in office (proto-Caps 1720–1738, Hats 1738–1765, Caps 1765–1769, Hats 1769–1772) reflected not a mandated alternation but the way elite networks, policy cleavages, and parliamentary dynamics produced recurring patterns of peaceful turnover. For Roberts, the era demonstrates how far a polity could advance toward structured, recurrent elite competition well before modern party politics emerged.

As [Rustow \(1955\)](#) notes, the Age of Liberty was Sweden’s first fully developed experiment in parliamentary rule and drew on a representative tradition extending back to the medieval Riksdag. It cultivated durable habits of deliberation, negotiation, and compromise—traits of a political culture marked by early, organized, and peaceful contestation. Although Gustav III’s coup of 1772 curtailed parliamentary authority, these norms proved resilient and were consciously revived in the constitutional settlement of 1809.

7. CONCLUSION

A long-standing thesis in political economy relates cooperation among rival elites to mechanisms for power alternation. [Schumpeter \(1942\)](#) prominently recast democracy from this perspective and advanced the minimalist conception of democracy, which has been playing a crucial role in the modern theories of democracy ([Przeworski, 1999](#)). Nonetheless, it remains unclear why governing elites would transfer power: they could instead secure cooperation of their rivals through making political concessions without relinquishing their power. Indeed, the practices of political concessions have been documented and thoroughly studied in the literature of authoritarian power sharing.

Toward resolving the puzzle, this paper has investigated the interplay among power alternation, political concessions, and elite cooperation in an environment without binding enforcement. Our analysis elucidates the roles of power alternation and political concessions. On the one hand, political concessions allocate resources within a period and expand the feasibility of power alternation. On the other hand, power alternation not only redistributes resources intertemporally, but also disciplines political concessions without interrupting the continuation of cooperation. Efficient self-enforcing arrangements involve both instruments whose interaction is essential for sustaining cooperation.

We have analyzed how this interplay shapes the endogenous distribution of power by characterizing three norms of power alternation in efficient self-enforcing arrangements of

sustained cooperation. These norms show how transfer of power is delayed as last resort, how power transitions are governed by an implicit protocol assigning maximal bargaining leverage to every inaugurating incumbent, and how every incumbent's ability to preserve its power through concessions is maximized throughout its incumbency. Together, they imply maximal power stability: incumbencies persist whenever feasible and power alternation occurs only when necessary to sustain cooperation. Our analysis thus provides a new perspective on the link between the stability of power and the sustainability of cooperation.

Finally, we have shown how the dynamics of power among elite groups exhibit history dependence and that an incumbency may feature hegemony or precarity. We have derived comparative statics on how the stability of power varies with aspects of the environment. In particular, power stability deteriorates as transferring power becomes more costly.

Our analysis demonstrates that power alternation and political concessions are closely linked instruments employed by rival elites to maintain their long-term relationships, and thus suggests that competing arguments about these instruments be synthesized. At the same time, while we focus on self-enforcing arrangements among elite groups, it is crucial to understand the interplay between these long-term relationships and the formal institutions (for instance, constitutions in [North and Weingast \(1989\)](#), courts in [Myerson \(2008\)](#), electoral processes in [Fearon \(2011\)](#), and parliaments in [Boix and Svolik \(2013\)](#)). Furthermore, as noted by [Dahl \(1971\)](#), practices of power alternation among elites laid the foundation of a stable mass democracy. However, there has been little systematic formalization of this important idea. In particular, how would the interaction between elite groups change when alternation of power also depends on the decisions of a strategic third party such as voters? Our paper is a first step toward these important directions of future research.

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